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ELECTRIC VEHICLE (EV) LOCKOUT APPARATUSES AND METHODS

Abstract

An electric vehicle (EV) lockout device including a clip and a housing. The housing having a base with a front side and a back side. A first flange extends from the front side of the base and is configured to engage with an EV charge port or EV charge plug. A clip support unit extending from the back side of the base. The clip support unit including a first side plate having a first top end, a first bottom end, and a first opening disposed between the first top end and the first bottom end, and a second side plate having a second top end, a second bottom end, and a first opening disposed between the second top end and the second bottom end. The second side plate being disposed substantially parallel to the first side plate, and the second opening being substantially aligned with the first opening.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] None.

FIELD OF DISCLOSURE

[0002] The present disclosure relates generally to electric vehicle lockout devices, and more specifically to electric vehicle lockout devices for a charge port of an electric vehicle and a charging plug for an electric vehicle charging station.

BACKGROUND

[0003] The number of electric vehicles being produced and driven has increased rapidly over the past decade. As a result, more electric vehicle (EV) charging stations have been built and installed to accommodate the growing number of electric vehicles.

[0004] Electric vehicles include a charge port that can be connected to a charge plug of the EV charging station to charge the battery of the electric vehicle. There are several types of charging ports and charging plugs used to charge an electric vehicle. Such ports and plugs include the Combined Charging System (CCS) Type 1 and CCS Type 2 charge ports and charge plugs, the J1772 charge port and charge plug, and the North American Charging Standard (NACS) or Tesla charge port and charge plug.

[0005] Currently, charge ports of electric vehicles and charge plugs of EV charge stations do not have an effective mechanism to lock the charge port or charge plug to prevent access. While some EV lockout devices exist that lock the socket housing of the electric vehicle to prevent someone from removing the charge port from the electric vehicle, the prior art devices do not necessarily prevent access to the charge port nor do they provide protection for charge plugs of EV charging stations. Thus, there exists a need for EV lockout devices that prevent access, namely, EV lockout devices that lock the charge port of the electric vehicle to prevent charging during maintenance or repairs of the electric vehicle and/or to prevent tampering with the charge port, and lock the charge plug of an EV charging station to prevent the charge plug from being used during maintenance or repairs and/or to prevent unauthorized use of the charge plug.

SUMMARY

[0006] The present apparatuses overcome many of the shortcomings and limitations of the prior art devices discussed above. The apparatuses and methods described herein include several EV lockout devices including multiple embodiments of housings for the EV lockout devices and multiple embodiments of clips for the EV lockout devices.

[0007] In one aspect, a housing for an electric vehicle (EV) lockout device is provided. The housing includes a base having a front side and a back side and a first flange extending from the front side of the base. The first flange is configured to engage with a first portion of the port housing of an EV charge port or a first portion of the plug housing of an EV charge plug. The housing also includes a clip support unit extending from the back side of the base. The clip support unit includes a first side plate having a first top end, a first bottom end that is opposite the first top end, and a first opening disposed between the first top end and the first bottom end. The clip support unit also includes a second side plate having a second top end, a second bottom end that is opposite the second top end, and a first opening disposed between the second top end and the second bottom end. The second side plate is disposed in a substantially parallel manner with respect to the first side plate, and the second opening is substantially aligned with the first opening.

[0008] In some instances, the first opening and the second opening are imparted with substantially the same shape and size. In some instances, the first opening and the second opening may be configured to align with a hole of a clip for an EV lockout device when the clip is attached to the housing for an EV lockout device. In some instances, the first opening and the second opening are configured to enable a shackle of a lock to pass through the first opening and the second opening when the lock is attached to the housing for an EV lockout device.

[0009] In other instances, the first flange includes a bottom flange portion. The housing for an EV

lockout device further includes a ridge that extends from the bottom flange portion of the first flange. The ridge is configured to engage with a channel in the first portion of the port housing of the EV charge port. In further instances, the housing for an EV lockout device further includes a flange channel that extends through the bottom flange portion of the first flange. The flange channel is configured to engage with a protrusion ridge of the plug housing of the EV charge plug. [0010] In some instances, the housing for an EV lockout device further includes a first vertical member disposed on the back side of the base and a second vertical member disposed on the back side of the base. The first vertical member is oriented in a substantially parallel manner with respect to the second vertical member. In some instances, the housing for an EV lockout device also includes a clip connector that extends between the first vertical member and the second vertical member. In some instances, the clip connector is configured to engage with a coupling member and a projection of a clip for an EV lockout device when the clip is attached to the housing for an EV lockout device.

[0011] In other instances, the housing for an EV lockout device includes a first support member connected to the first top end of the first side plate, and a second support member connected to the second top end of the second side plate. The first vertical member extends upwardly from the first support member and the second vertical support extends upwardly from the second support member. In some instances, the first support member and the second support member are configured to engage with a first support leg and a second support leg, respectively, of a clip for an EV lockout device when the clip is attached to the housing for an EV lockout device.

[0012] In some instances, the housing for an EV lockout device includes a second flange extending from the front side of the base. The second flange is disposed below the first flange and is configured to engage with a second portion of the port housing of the EV charge port or the second portion of the plug housing of the EV charge plug. In some instances, a ridge is disposed between the first flange and the second flange. The ridge is configured to engage with a channel in the first portion of the port housing of the EV charge port. In other instances, a flange channel is disposed between the first flange and the second flange. The flange channel is configured to engage with a protrusion ridge of the first portion of the plug housing of the EV charge port.

[0013] In some instances, the housing for an EV lockout device includes a first guide rail and a second guide rail disposed along a top flange portion of the first flange. The first guide rail and the second guide rail are configured to engage with a first guide member and a second guide member, respectively, of a clip for an EV lockout device when the clip is attached to the housing for an EV lockout device.

[0014] In another aspect, a clip for an electric vehicle (EV) lockout device is provided. The clip includes a first elongated member having a first end and a second end that is opposite the first end. The clip includes a second elongated member having a top portion and a bottom portion opposite the top portion. The second elongated member is disposed substantially perpendicular to the first elongated member. The clip also includes a platform portion that is connected to the second end of the first elongated member. A first support leg and a second support leg extend downwardly from the platform portion. The second support leg is disposed in a substantially parallel orientation with respect to the first support legs. In some instances, the first support leg extends downwardly from a front end of the platform portion on a first platform side of the platform portion, and the second support leg extends downwardly from the front end of the platform portion on a second platform side of the platform portion.

[0015] In some instances, the second elongated member extends downwardly from the platform portion. In other instances, the second elongated member extends upwardly from a top side of the first elongated member.

[0016] In some instances, the clip for an EV lockout device includes a hole disposed between the top portion and the bottom portion of the second elongated member. In some instances, the hole is configured to substantially align with a first opening and a second opening of a clip support unit of

a housing for an EV lockout device when the clip for an EV lockout device is attached to the housing for an EV lockout device. In some instances, the first opening and the second opening of the clip support unit for an EV lockout device and the hole of the clip for an EV lockout device are configured to enable a shackle of a lock to pass through the first opening, the hole, and the second opening, when the clip is attached to a housing for an EV lockout device and a lock is attached to the EV lockout device.

[0017] In some instances, the first support leg and the second support leg each extend downwardly from the platform portion at an angle. In some instances, the first support leg and the second support leg are configured to compress such that the angle between the platform portion and the first support leg or the second support leg decreases. In some instances, the angle is about 45 degrees. In some instances, the first support leg and the second support leg are configured to engage with a first support member and a second support member, respectively, of a clip support unit of a housing for an EV lockout device when the clip for an EV lockout device is attached to the housing for an EV lockout device. In other instances, the first support leg and the second support leg are configured to engage with a bottom clip support of a housing for an EV lockout device when the clip for an EV lockout device is attached to the housing for an EV lockout device.

[0018] In some instances, the clip for an EV lockout device includes a hook portion that extends downwardly from the first end of the first elongated member. The hook portion is configured to engage with a locking mechanism of an EV charge port or an EV charge plug. In other instances, the clip for an EV lockout device includes a raised member on the first end of the first elongated member. The raised member is configured to engage with a locking mechanism of an EV charge port or an EV charge plug.

[0019] In some instances, the first elongated member includes a first side and second side that is substantially parallel to the first side. The first side and the second side extend between the first end and the second end of the first elongated member. In some instances, a first guide member extends downwardly from the first side of the first elongated member, and a second guide member extends downwardly from the second side of the first elongated member. The first guide member and the second guide member are configured to engage with a first guide rail and a second guide rail, respectively, of a housing for an EV lockout device when the clip for an EV lockout device is attached to the housing for an EV lockout device.

[0020] In yet another aspect, a method of using an electric vehicle (EV) lockout device is provided. The EV lockout device includes a clip and a housing. The clip includes a first elongated member having a first end and a second end that is opposite the first end. The clip also includes a second elongated member disposed perpendicular to the first elongated member. A hole is disposed on the second elongated member. The clip further includes a platform portion connected to the second end of the first elongated member.

[0021] The method includes the step of attaching the clip to the housing such that the first elongated member extends over a first flange of the housing and the hole is in substantial alignment with a first opening of the housing and a second opening of the housing.

[0022] The method includes the step of depressing the platform portion of the clip to move the clip from a first clip position in which the clip is in a neutral position to a second clip position in which the first end of the first elongated member is raised. The method includes the steps of engaging the first end of the clip with a locking mechanism of an EV charge port or an EV charge plug and mating the first flange of the housing with a first portion of a port housing of the EV charge port or a first portion of the plug housing of the EV charge plug. In some instances, the method includes the step of mating a second flange of the housing with a second portion of a port housing of the EV charge port or a second portion of the plug housing of the EV charge plug.

[0023] The method includes the step of releasing the platform portion of the clip to move the clip from the second clip position to the first clip position.

[0024] The method further includes the step of attaching a lock to the EV lockout device by

threading a shackle of a lock through the first opening of the housing, the hole of the second elongated member of the clip, and the second opening of the housing, and attaching the shackle to a body of the lock.

[0025] These and other aspects and advantages of the present disclosure will become apparent to those skilled in the art after considering the following detailed description in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0026] FIG. 1 is a partially exploded isometric view of a lock, a first embodiment of an electric vehicle (EV) lockout device, and a first charge port for an electric vehicle;

[0027] FIG. 2A is an isometric view of a top, back, and right side of a housing for the first embodiment of the EV lockout device of FIG. 1;

[0028] FIG. 2B is a front plan view of the housing of FIG. 2A;

[0029] FIG. 2C is a left side elevational view of the housing of FIG. 2A, the right side view being a mirror image thereof;

[0030] FIG. 2D is a rear plan view of the housing of FIG. 2A;

[0031] FIG. 3A is an isometric view of a top, back, and right side of a first embodiment of a clip for use with one or more of the first embodiment of the EV lockout device of FIG. 1, the third embodiment of the EV lockout device of FIG. 11, and the seventh embodiment of the EV lockout device of FIG. 29;

[0032] FIG. 3B is a left side elevational view of the first embodiment of the clip of FIG. 3A, the right side view being a mirror image thereof;

[0033] FIG. 3C is a front plan view of the first embodiment of the clip of FIG. 3A;

[0034] FIG. 4A is an isometric view of a top, back, and right side of the first embodiment of the EV lockout device of FIG. 1 including the housing of FIGS. 2A-2D in combination with the first embodiment of the clip of FIGS. 3A-3C;

[0035] FIG. 4B is a rear plan view of the first embodiment of the EV lockout device of FIG. 4A;

[0036] FIG. 4C is a cross-sectional view of the first embodiment of the EV lockout device taken generally along the lines 4C-4C of FIG. 4A;

[0037] FIG. 5 is an isometric view of the first embodiment of the EV lockout device of FIGS. 1 and 4A-4C in use with the lock and the first charge port for an electric vehicle of FIG. 1;

[0038] FIG. 6 is a partially exploded isometric view of a lock, a second embodiment of an EV lockout device, and a first charge plug for an electric vehicle;

[0039] FIG. 7A is an isometric view of a top, back, and right side of a housing for the second embodiment of the EV lockout device of FIG. 6;

[0040] FIG. 7B is a front plan view of the housing of FIG. 7A;

[0041] FIG. 7C is a left side elevational view of the housing of FIG. 7A, the right side view being a mirror image thereof;

[0042] FIG. 7D is a rear plan view of the housing of FIG. 7A;

[0043] FIG. 8A is an isometric view of a top, back, and right side of a second embodiment of a clip for use with the second embodiment of the EV lockout device of FIG. 6 and the fourth embodiment of the EV lockout device of FIG. 15;

[0044] FIG. 8B is a left side elevational view of the second embodiment of the clip of FIG. 8A, the right side view being a mirror image thereof;

[0045] FIG. 8C is a front plan view of the second embodiment of the clip of FIG. 8A;

[0046] FIG. 9A is an isometric view of a top, back, and right side of the second embodiment of the EV lockout device of FIG. 6 including the housing of FIGS. 7A-7D in combination with the second

embodiment of the clip of FIGS. 8A-8C;

[0047] FIG. 9B is rear plan view of the second embodiment of the EV lockout device of FIG. 9A;

[0048] FIG. 9C is a cross-sectional view of the second embodiment of the EV lockout device taken generally along the lines 9C-9C of FIG. 9A;

[0049] FIG. 10 is an isometric view of the second embodiment of the EV lockout device of FIGS. 6 and 9A-9C in use with the lock and the first charge plug for an electric vehicle of FIG. 6;

[0050] FIG. 11 is a partially exploded isometric view of a lock, a third embodiment of an EV lockout device, and a second charge port for an electric vehicle;

[0051] FIG. 12A is an isometric view of a top, back, and right side of a housing for the third embodiment of the EV lockout device of FIG. 11;

[0052] FIG. 12B is a front plan view of the housing of FIG. 12A;

[0053] FIG. 12C is a left side elevational view of the housing of FIG. 12A, the right side view being a mirror image thereof;

[0054] FIG. 12D is a rear plan view of the housing of FIG. 12A;

[0055] FIG. 13A is an isometric view of a top, back, and right side of the third embodiment of the EV lockout device of FIG. 11 including the housing of FIGS. 12A-12D in combination with the first embodiment of the clip of FIGS. 3A-3C;

[0056] FIG. 13B is a rear plan view of the third embodiment of the EV lockout device of FIG. 13A;

[0057] FIG. 13C is a cross-sectional view of the third embodiment of the EV lockout device taken generally along the lines 13C-13C of FIG. 13A;

[0058] FIG. 14 is an isometric view of the third embodiment of the EV lockout device of FIGS. 11 and 13A-13C in use with the lock and the second charge port for an electric vehicle of FIG. 11;

[0059] FIG. 15 is a partially exploded isometric view of a lock, a fourth embodiment of an EV lockout device, and a second charge plug for an electric vehicle;

[0060] FIG. 16A is an isometric view of a top, back, and right side of a housing for the fourth embodiment of the EV lockout device of FIG. 15;

[0061] FIG. 16B is a front plan view of the housing of FIG. 16A;

[0062] FIG. 16C is a left side elevational view of the housing of FIG. 16A, the right side view being a mirror image thereof;

[0063] FIG. 16D is a rear plan view of the housing of FIG. 16A;

[0064] FIG. 17A is an isometric view of a top, back, and right side of the fourth embodiment of the EV lockout device of FIG. 15 including the housing of FIGS. 16A-16D in combination with the second embodiment of the clip of FIGS. 8A-8C;

[0065] FIG. 17B is a rear plan view of the fourth embodiment of the EV lockout device of FIG. 17A;

[0066] FIG. 17C is a cross-sectional view of the fourth embodiment of the EV lockout device taken generally along the lines 17C-17C of FIG. 17A;

[0067] FIG. 18 is an isometric view of the fourth embodiment of the EV lockout device of FIGS. 15 and 17A-17C in use with the lock and the second charge plug for an electric vehicle of FIG. 15;

[0068] FIG. 19 is a partially exploded isometric view of a lock, a fifth embodiment of an EV lockout device, and a third charge plug for an electric vehicle;

[0069] FIG. 20A is an isometric view of a top, back, and right side of a housing for the second embodiment of the EV lockout device of FIG. 19;

[0070] FIG. 20B is a front plan view of the housing of FIG. 20A;

[0071] FIG. 20C is a left side elevational view of the housing of FIG. 20A, the right side view being a mirror image thereof;

[0072] FIG. 20D is a rear plan view of the housing of FIG. 20A;

[0073] FIG. 21A is an isometric view of a top, front, and right side of a third embodiment of a clip for use with the fifth embodiment of the EV lockout device of FIG. 19;

[0074] FIG. 21B is a left side elevational view of the third embodiment of the clip of FIG. 21A, the

right side view being a mirror image thereof;

[0075] FIG. 21C is a front plan view of the third embodiment of the clip of FIG. 21A;

[0076] FIG. 22A is an isometric view of a top, back, and right side of the fifth embodiment of the EV lockout device of FIG. 19 including the housing of FIGS. 20A-20D in combination with the third embodiment of the clip of FIGS. 21A-21C;

[0077] FIG. 22B is a rear plan view of the fifth embodiment of the EV lockout device of FIG. 22A;

[0078] FIG. 22C is a cross-sectional view of the fifth embodiment of the EV lockout device taken generally along the lines 22C-22C of FIG. 22A;

[0079] FIG. 23 is an isometric view of the fifth embodiment of the EV lockout device of FIGS. 19 and 22A-22C in use with the lock and the third charge plug for an electric vehicle of FIG. 19;

[0080] FIG. 24 is a partially exploded isometric view of a lock, a sixth embodiment of an EV lockout device, and a third charge port for an electric vehicle;

[0081] FIG. 25A is an isometric view of a top, back, and right side of a housing for the sixth embodiment of the EV lockout device of FIG. 24;

[0082] FIG. 25B is a left side elevational view of the housing of FIG. 25A, the right side view being a mirror image thereof;

[0083] FIG. 25C is a front plan view of the housing of FIG. 25A;

[0084] FIG. 25D is a rear plan view of the housing of FIG. 25A;

[0085] FIG. 26A is an isometric view of a top, front, and right side of a fourth embodiment of a clip for use with the sixth embodiment of the EV lockout device of FIG. 24;

[0086] FIG. 26B is a left side elevational view of the fourth embodiment of the clip of FIG. 26A, the right side view being a mirror image thereof;

[0087] FIG. 26C is a front plan view of the fourth embodiment of the clip of FIG. 26A;

[0088] FIG. 27A is an isometric view of a top, back, and right side of the sixth embodiment of the EV lockout device of FIG. 24 including the housing of FIGS. 25A-25D in combination with the fourth embodiment of the clip of FIGS. 26A-26C;

[0089] FIG. 27B is a bottom view of the housing of FIG. 27A;

[0090] FIG. 27C is a rear plan view of the sixth embodiment of the EV lockout device of FIG. 27A;

[0091] FIG. 27D is a cross-sectional view of the sixth embodiment of the EV lockout device taken generally along the lines 27D-27D of FIG. 27A;

[0092] FIG. 28 is an isometric view of the sixth embodiment of the EV lockout device of FIGS. 24 and 27A-27D in use with the lock and the third charge port for an electric vehicle of FIG. 24;

[0093] FIG. 29 is a partially exploded isometric view of a lock, a seventh embodiment of an EV lockout device, and a fourth charge plug for an electric vehicle;

[0094] FIG. 30A is an isometric view of a top, back, and left side of a housing for the seventh embodiment of the EV lockout device of FIG. 29;

[0095] FIG. 30B is a front plan view of the housing of FIG. 30A;

[0096] FIG. 30C is a right side elevational view of the housing of FIG. 30A, the left side view being a mirror image thereof;

[0097] FIG. 30D is a rear plan view of the housing of FIG. 30A;

[0098] FIG. 31A is an isometric view of a top, back, and left side of the seventh embodiment of the EV lockout device of FIG. 29 including the housing of FIGS. 30A-30D in combination with the first embodiment of the clip of FIGS. 3A-3C;

[0099] FIG. 31B is a rear view of the seventh embodiment of the EV lockout device of FIG. 31A;

[0100] FIG. 31C is a cross-sectional view of the seventh embodiment of the EV lockout device taken generally along the lines 31C-31C of FIG. 31A;

[0101] FIG. 32 is an isometric view of the seventh embodiment of the EV lockout device of FIGS. 29 and 31A-31C in use with the lock and the fourth charge plug for an electric vehicle of FIG. 29;

and

[0102] FIG. 33 is a flowchart of a method 1200 of using the first, second, third, fourth, fifth, and sixth embodiments of the EV lockout devices of FIGS. 1, 6, 11, 15, 19, 24, and 29, respectively. [0103] While the disclosure is susceptible to various modifications and alternative forms, a specific embodiment thereof is shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description presented herein are not intended to limit the disclosure to the particular embodiment disclosed, but to the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

DETAILED DESCRIPTION

[0104] Before any embodiments are described in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings, which is limited only by the claims that follow the present disclosure. The disclosure is capable of other embodiments, and of being practiced, or of being carried out, in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “attached,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, attachments, supports, and couplings. Further, “connected,” “attached,” and “coupled” are not restricted to physical or mechanical connections, attachments, or couplings.

[0105] The following description is presented to enable a person skilled in the art to make and use embodiments of the disclosure. Various modifications to the illustrated embodiments will be readily apparent to those skilled in the art, and the generic principles herein can be applied to other embodiments and applications without departing from embodiments of the disclosure. Thus, embodiments of the disclosure are not intended to be limited to embodiments shown but are to be accorded the widest scope consistent with the principles and features disclosed herein. The following detailed description is to be read with reference to the figures, in which like elements in different figures have like reference numerals. Furthermore, directional terminology is utilized herein (e.g., front, back, right, left, upwards, downwards, etc.) to describe components or features in relation to one another and is not intended to be limiting. Skilled artisans will recognize the examples provided herein have many useful alternatives and fall within the scope of embodiments of the disclosure.

[0106] Additionally, while the following discussion may describe features associated with specific devices or embodiments, it is understood that additional devices and/or features can be used with the described systems and methods, and that the discussed devices and features are used to provide examples of possible embodiments, without being limited.

[0107] The embodiments of the EV lockout devices disclosed herein may include a housing and an embodiment of a clip. The clip may be placed on, attached to, or otherwise engaged with the housing. Once the clip is placed on the housing, the EV lockout device may be placed on and secured to an EV charge port or EV charge plug with a locking device.

[0108] The embodiments of the EV lockout devices disclosed herein provide numerous benefits. For example, the EV lockout devices can protect the charge port of the electric vehicle or the charge plug of an EV charge station while maintenance or repairs are performed on the electric vehicle or the EV charging station. Locking the charge port of an electric vehicle with an EV lockout device may also prevent the electric vehicle from being charged while maintenance or repairs are being performed on the electric vehicle. In some instances, the EV lockout device can be left on the charge port when the charge port cover is closed, which may prevent the charge port from being tampered with while the electric vehicle is parked or being stored.

[0109] In addition, locking the charge plug of an EV charge station with an EV lockout device can prevent the charge plug of an EV charging station from being used if the charge plug is disabled or unsafe to use or maintenance or repairs are being performed. Furthermore, locking the charge plug of an EV charging station with an EV lockout device may prevent unauthorized users from using the charge plug.

[0110] Furthermore, the unique EV lockout device disclosed herein are designed for easy installation on and removal from the charge port of an electric vehicle or the charge plug of an EV charging station (e.g., single-handed installation and/or removal). Some embodiments of the EV lockout devices include clips that are designed to lift up and snap down over a portion of the housing of the EV charge port or the EV charge plug just by pushing the EV lockout device onto the housing of the EV charge port or the EV charge plug. Other embodiments of the EV lockout devices include clips that are designed to simply snap in place just by pushing the EV lockout devices onto or into the housing of the EV charge port or the EV charge plug. The platform portion of the clips may then be depressed using a finger or thumb to disengage the clips from the EV charge port or the EV charge plug for easy removal of the EV lockout device.

[0111] FIG. 1 depicts a partially exploded view of a lock **10**, a first embodiment of an EV lockout device **100**, and a first electric vehicle (EV) charge port **20**. The lock **10** may be provided in the form of a body **12** and a shackle **14** that securely attaches to the body **12** when in a locked position as shown in FIG. 1. The shackle **14** may be defined by a first end portion **16** and a second end portion **18**. One or both of the first end portion **16** and the second end portion **18** may disengage from the body **12** when the lock **10** is in an unlocked position. In one embodiment, the first end portion **16** of the shackle **14** may disengage from the body **12**, while the second end portion **18** remains fastened or otherwise attached to the body **12**, when in the unlocked position. The first end portion **16** may be movable relative to the body **12**, when in the unlocked position, such that the shackle **14** can be threaded through an opening or other aperture in an object to be secured (e.g., the EV lockout device **100**) before the first end portion **16** of the shackle **14** is reattached to the body **12**, when in the locked position (see, e.g., FIG. 5).

[0112] The first EV charge port **20** may be disposed inside of, on, or otherwise associated with an electric vehicle (not shown). In some embodiments, the first EV charge port **20** may be provided in the form of a Combined Charging System (CCS) Type I socket.

[0113] The first EV charge port **20** may include a port housing **22** having a locking mechanism aperture **24**, a first protrusion aperture **26**, and a second protrusion aperture **28**. The first EV charge port **20** may include a locking mechanism **30** disposed within the locking mechanism aperture **24**, which may include a locking member **32** having an inner side **33**.

[0114] The first EV charge port **20** may further include a first protrusion **34** disposed within the first protrusion aperture **26**. A first set of terminals receptors **36** may be disposed within the first protrusion **34**.

[0115] The first protrusion **34** may be imparted with a first protrusion shape. In some embodiments, the first protrusion shape may be substantially cylindrical with a substantially circular cross-section. In other embodiments, the first protrusion **34** may be imparted with other shapes, for example, a rectangular or square shape, a triangular shape, a trapezoidal shape, and the like, a combination of shapes, or an irregular shape.

[0116] The first protrusion **34** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the first protrusion **34** may include a diameter, a circumference, and/or a depth. In some embodiments, the diameter of the first protrusion **34** may be about 33.9 millimeters (mm). In some embodiments, the depth of the first protrusion **34** may be about 35.5 mm.

[0117] The first EV charge port **20** may include a second protrusion **38** disposed within the second protrusion aperture **28**. A second set of terminal receptors **40** may be disposed within the second protrusion **38**.

[0118] The second protrusion **38** may be imparted with a second protrusion shape. In some embodiments, the second protrusion shape may be substantially cylindrical with a substantially elliptical cross-section. In other embodiments, the second protrusion **38** may be imparted with other shapes, for example, a rectangular or square shape, a triangular shape, a trapezoidal shape, and the like, a combination of shapes, or an irregular shape.

[0119] The second protrusion **38** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the second protrusion **38** may include one or more diameters, a circumference, and/or a depth. One or more of the dimensions within the set of dimensions of the second protrusion **38** may be the same or different from one or more of the dimensions within the set of dimensions for the first protrusion **34**. In some embodiments, the depth of the first protrusion **34** may be the same as the depth of the second protrusion **38**.

[0120] In some embodiments, a first diameter (or width) of the second protrusion **38** may be about 51.34 mm, and a second diameter (or a height) of the second protrusion **38** may be about 25.34 mm. In some embodiments, the depth of the second protrusion **38** may be about 35.5 mm.

[0121] In some embodiments, the locking mechanism aperture **24** may be disposed above the first protrusion aperture **26**, and the first protrusion aperture **26** may be disposed above the second protrusion aperture **28**. A divider **42** may be disposed between the locking mechanism aperture **24** and the first protrusion aperture **26**. A channel **44** may be disposed between the first protrusion aperture **26** and the second protrusion aperture **28**.

[0122] The channel **44** may be defined by a first channel side **46a** and the second channel side **46b**. In some embodiments, the channel **44** may be imparted with a substantially rectilinear shape, although other shapes may be used. The channel **44** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the channel **44** may include a width, a height, and a depth. For example, in some embodiments, the width of the channel **44** may be about 9.1 mm. In some embodiments, the height of the channel **44** may be about 3.48 mm. In some embodiments, the depth of the channel **44** may be about 35.5 mm.

[0123] The depth of the channel **44** may be the same or different than the depth of the first protrusion **34** or the depth of the second protrusion **38**. In some embodiments, the depths of the channel **44**, the first protrusion **34**, and the second protrusion **38** may be substantially the same, or the same.

[0124] Turning to FIGS. 2A-2D, the first embodiment of the EV lockout device **100** may include a housing **102**. The housing **102** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the housing **102** may be made of acrylonitrile butadiene styrene (ABS) polymer plastic. In some embodiments, the housing **102** may be formed using injection molding or a 3D printer.

[0125] The housing **102** may include a base **104** defined by a front side **106** and a back side **108**. The base **104** may also have a top portion **110** and a bottom portion **112**, opposite the top portion **110**. In some embodiments, the top portion **110** terminates at a top edge **111**. The base **104** may have an outer edge **114** that defines a shape of the base **104**. In some embodiments, the shape of the base **104** may be substantially a square, a rectangle, a circle, an oval, a triangle, a trapezoid, and the like. In some embodiments, the shape of the base **104** may be imparted with round edges as shown in FIGS. 2A, 2B, and 2D. Combinations of shapes and irregular shapes are also contemplated.

[0126] In some embodiments, the base **104** may be imparted with a substantially trapezoidal shape as shown in FIGS. 2B and 2D where a top length **116** dimension of the top portion **110** is less than a bottom length **118** dimension of the bottom portion **112**. In some embodiments, the top length **116** of the top portion **110** may be 40.23 mm, or about 40.23 mm, and the bottom length **118** of the bottom portion **112** may be 65 mm, or about 65 mm.

[0127] The base **104** may be imparted with a base height **120**. In some embodiments, the base height **120** may be 95.32 mm, or about 95.32 mm. The base **104** also may be imparted with a base thickness (not shown). In some embodiments, the base thickness may be 3 mm, or about 3 mm.

[0128] The top portion **110** of the base **104** may be interrupted by a divot **122** disposed in a center area of the top portion **110**. The divot **122** may extend downwardly from the top edge **111** and be defined by a first divot side **123a**, a second divot side **123b**, and a bottom divot side **123c**. In some embodiments, the divot **122** may be imparted with a “C” shape or a “U” shape as shown in FIGS. 2B and 2D.

[0129] The front side **106** of the base **104** may include a first flange **124** and a second flange **126** protruding outwardly therefrom. The first flange **124** may have a top flange portion **125** and a bottom flange portion **127**. In some embodiments, the first flange **124** may be disposed above the second flange **126** and below the divot **122** of the top portion **110**. The first flange **124** may be connected to the second flange **126** by a ridge **128**. In some embodiments, the ridge **128** may extend downwardly from a center area of the bottom flange portion **127**.

[0130] The first flange **124** may have a first flange wall **130** that is imparted with a first flange thickness. In some embodiments, the first flange thickness may be 2.5 mm, or about 2.5 mm.

[0131] The first flange wall **130** may define a first flange shape. The first flange shape may be any size and shape suitable to engage with the first protrusion **34** when the EV lockout device **100** is placed on or otherwise in communication with the first EV charge port **20**. In some embodiments, the first flange shape may be the same or substantially the same as the shape of the first protrusion **34** of the first EV charge port **20**. In some embodiments, the first flange shape may be substantially cylindrical with a substantially circular cross-section.

[0132] The first flange **124** may be imparted with a first set of dimensions. In some embodiments, the first set of dimensions of the first flange **124** may be complimentary to the set of dimensions for the first protrusion **34** of the first EV charge port **20** so that the first flange **124** can be placed onto and fit over the first protrusion **34**. In some instances, a friction fit between the first flange wall **130** and the first protrusion **34** may prevent the housing **102** from disengaging from the first EV charge port **20**.

[0133] In some embodiments, the first set of dimensions of the first flange **124** may include a first flange diameter **132** (see FIG. 2B), a first flange circumference (not shown), and a first flange depth **134** (see FIG. 4C). In some embodiments, the first flange diameter **132** may be the same as or greater than the diameter of the first protrusion **34** of the first EV charge port **20**. For example, in some embodiments, the first flange diameter **132**, may be 38 mm, or about 38 mm, and the first flange circumference may be 119.41 mm, or about 119.41 mm.

[0134] The first flange depth **134** may be the same as or different from the depth of the first protrusion **34** of the first EV charge port **20**. In some embodiments, the first flange depth **134** is the same as, substantially the same as or less than the depth of the first protrusion **34** of the first EV charge port **20**. For example, in some embodiments, the first flange depth **134** may be 30 mm, or about 30 mm.

[0135] The second flange **126** may have a second flange wall **136** that may be imparted with a second flange thickness. In some embodiments, the second flange thickness may be 2.5 mm, or about 2.5 mm.

[0136] The second flange wall **136** may define a second flange shape. The second flange shape may be any size and shape suitable to engage with the second protrusion **38** when the EV lockout device **100** is placed on or otherwise in communication with the first EV charge port **20**. In some embodiments, the second flange shape may be the same or substantially the same as the second protrusion **38** of the first EV charge port **20**. In some embodiments, the second flange shape may be substantially cylindrical with a substantially elliptical cross-section.

[0137] The second flange **126** may be imparted with a second set of dimensions. In some embodiments, the second set of dimensions of the second flange **126** may be complimentary to the set of dimensions for the second protrusion **38** of the first EV charge port **20** so that the second flange **126** can be placed onto and fit over the second protrusion **38**. In some instances, a friction fit between the second flange wall **136** and the second protrusion **38** may prevent the housing **102**

from disengaging from the first EV charge port **20**.

[0138] In some embodiments, the second set of dimensions of the second flange **126** may include a second flange diameter **138**, a second flange height **140** (see FIG. 2B), a second flange circumference (not shown), and a second flange depth **142** (see FIG. 4C). In some embodiments, the second flange diameter **138** may be the same as or greater than the first diameter of the second protrusion **38** of the first EV charge port **20**. In some embodiments, the second flange height **140** may be the same as or greater than the second diameter of the second protrusion **38** of the first EV charge port **20**. For example, in some embodiments, the second flange diameter (or width) **138** may be 53.51 mm, or about 53.51 mm; and, the second flange height **140** may be 27.11 mm, or about 27.11 mm. The second flange circumference may be 137.98 mm, or about 137.98 mm.

[0139] The second flange depth **142** may be the same as or different from the depth of the second protrusion **38** of the first EV charge port **20**. In some embodiments, the second flange depth **142** may be the same as, substantially the same as, or less than the depth of the second protrusion **38** of the first EV charge port **20**. In other embodiments, the second flange depth **142** may be the same as the first flange depth **134**. For example, in some embodiments, the second flange depth **142** may be 30 mm, or about 30 mm.

[0140] The ridge **128** may be imparted with a size and shape that is designed to engage with the channel **44** of the first EV charge port **20**. In some embodiments, the ridge **128** may be imparted with a substantially rectangular shape, although other shapes may be used that can engage with the channel **44** of the first EV charge port **20**.

[0141] In some embodiments, the ridge **128** may be defined by a ridge width **144** (see FIG. 2B), a ridge height **146**, and a ridge depth **148** (see FIG. 2C). The ridge width **144** may be the same as or less than the width of the channel **44** of the first EV charge port **20**. In some embodiments, the ridge width **144** may be 6.49 mm, or about 6.49 mm. The ridge height **146** may be the same or greater than the height of the channel **44** of the first EV charge port **20**. In some embodiments, the ridge height **146** may be 4.55 mm, or about 4.55 mm.

[0142] The ridge depth **148** may be the same as or different than the depth of the channel **44** of the first EV charge port **20**. In addition, the ridge depth **148** may be the same as or different than the first flange depth **134** or the second flange depth **142**. In some embodiments, the ridge depth **148**, the first flange depth **134**, and the second flange depth **142** may be the same. In some embodiments, the ridge depth **148** may be 30 mm, or about 30 mm.

[0143] The back side **108** of the base **104** may include a clip support unit **150** protruding outwardly therefrom. The clip support unit **150** may be defined by a bottom support **152**, a first side **154a** and a second side **154b**. In some embodiments, the first side **154a** may be substantially parallel to the second side **154b**. In some embodiments, the second side **154b** may be a mirror image of the first side **154a**.

[0144] The first side **154a** may include a first side plate **156a** having a first bottom end **158a** and a first top end **160a**. The first side plate **156a** may be attached to the bottom support **152** on the first bottom end **158a**. The second side **154b** may include a second side plate **156b** having a second bottom end **158b** and a second top end **160b**. The second side plate **156b** may be attached to the bottom support **152** on the second bottom end **158b**.

[0145] The bottom support **152** may connect the first bottom end **158a** of the first side plate **156a** to the second bottom end **158b** of the second side plate **156b**. A gap **162** is formed between the first side plate **156a** and the second side plate **156b** as shown in FIG. 2D.

[0146] A first hole **164a** may be disposed in and extend entirely through the first side plate **156a** between the first top end **160a** and the first bottom end **158a**. A second hole **164b** may also be disposed in and extend entirely through the second side plate **156b** between the second top end **160b** and the second bottom end **158b**. The second hole **164b** may be disposed in the second side plate **156b** at the same location as the first hole **164a** is disposed in the first side plate **156a** so that the first hole **164a** and the second hole **164b** are substantially aligned. The first and second holes

164a, **164b** may be any size and shape suitable for a shackle or shank of a lock (e.g., shackle **14** of the lock **10**) to pass through. In some embodiments, the first and second holes **164a**, **164b** may be imparted with a substantially oval shape. In other embodiments, the first and second holes **164a**, **164b** may be imparted with a substantially circular shape.

[0147] The first side **154a** may further include a first clip support **166a** on the first top end **160a** and the second side **154b** may include a second clip support **166b** on the second top end **160b**. A first vertical support **168a** may extend from the first clip support **166a** toward the top edge **111** of the base **104** and a second vertical support **168b** may extend from the second clip support **166b** toward the top edge **111** of the base **104**. In some embodiments, the first vertical support **168a** and the second vertical support **168b** may be mirror images of each other. The first and second vertical supports **168a**, **168b** may also be disposed on the back side **108** of the housing **102** adjacent the divot **122** such that the first vertical support **168a** is disposed adjacent the first divot side **123a** of the divot **122** and the second vertical support **168b** is disposed adjacent the second divot side **123b** of the divot **122**.

[0148] A clip connector **170** may extend between a first top **172a** of the first vertical support **168a** and a first top **172b** of the second vertical support **168b** thereby connecting the first vertical support **168a** to the second vertical support **168b**. The clip connector **170** may also extend across the divot **122** in such a manner that an opening **174** is defined by the first divot side **123a**, the second divot side **123b**, the bottom divot side **123c**, and the clip connector **170**. In some embodiments, the clip connector **170** may be imparted with a cylindrical or rod shape.

[0149] The clip support unit **150** may be designed to engage with one or more embodiments of a clip (e.g., **200**, **400**, **800**, **1000**) disclosed herein. In some embodiments, the first and second clip supports **166a**, **166b** may be designed to engage or otherwise be in communication with first and second support legs (e.g., **222a**, **222b**), respectively, of a clip (e.g., **200**). The clip connector **170** may be designed to engage with a coupling member (e.g., **216**) of the clip such that the coupling member can rotate about the clip connector **170** in order to raise and/or lower a first end (e.g., **204**) of a first elongated member (e.g., **202**) of the clip. In addition, the gap **162** disposed between the first and second side plates **156a**, **156b** may be designed to allow a second elongated member (e.g., **228**) of the clip to move in a substantially upwardly or downwardly direction between the first and second side plates **156a**, **156b**. Furthermore, the first and second holes **164a**, **164b** of the clip support unit **150** may be designed to substantially align with a lock through hole (e.g., **236**) of the clip to enable the shackle of a lock (e.g., shackle **14** of lock **10**) to pass through the housing **102** and clip and secure same to an EV charge port (e.g., **20**) or an EV charge plug.

[0150] The first embodiment of the EV lockout device **100** may include a first embodiment of a clip **200** designed for use therewith. The clip **200** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the clip **200** may be made of polypropylene. In some embodiments, the clip **200** may be formed using injection molding or a 3D printer.

[0151] As shown in FIGS. **3A** and **3B**, the clip **200** may be provided in the form of a body having a first elongated member **202** defining a first end **204** and a second end **206**, opposite the first end **204**. The first elongated member **202** may have a first elongated side **207a** and a second elongated side **207b**, whereby the first and second elongated sides **207a**, **207b** may extend between the first end **204** and the second end **206** of the first elongated member **202**. The first elongated member **202** may further be defined by a top side **208** and a bottom side **210**.

[0152] The elongated member **202** may terminate at a hook portion **212** that extends downwardly from the bottom side **210** at the first end **204**. In some embodiments, the hook portion **212** may be designed to engage with the locking mechanism **30** of the first EV charge port **20** of FIG. **1** (or the locking mechanism **30'** of the second EV charge port **20'** of FIG. **11** or the locking mechanism **54''** of the fourth EV charge plug **50''** of FIG. **29**, discussed below). In some embodiments, the hook portion **212** may be imparted with a substantially triangular cross-sectional profile. In other

embodiments, the hook portion **212** may be imparted with a different shape.

[0153] The first elongated member **202** may further include a first rail **214a** and a second rail **214b** that extend along the top side **208** of the first elongated member **202**. The first rail **214a** may be disposed on and be coextensive with the first elongated side **207a** and the second rail **214b** may be disposed on and be coextensive with the second elongated side **207b**. In some embodiments, the first and second rails **214a**, **214b** may provide structural support or additional rigidity to the first elongated member **202**.

[0154] The first elongated member **202** may further include a coupling member **216** protruding downwardly from the bottom side **210** thereof. The coupling member **216** may be disposed on the bottom side **210** of the first elongated member **202** adjacent the second end **206**. In some embodiments, the bottom side **210** of the first elongated member **202** together with the coupling member **216** may define a “C” or “U” shape nook **217** as shown in FIG. 3B. The coupling member **216** may include a projection **218** on an end portion of the coupling member **216**. In some embodiments, the projection **218** may extend upwardly toward the bottom side **210** of the first elongated member **202**. In some embodiments, the nook **217** and the projection **218** may be designed to keep the clip **200** engaged with the housing **102** when placed thereon (see, e.g., FIG. 4C), or to keep the clip **200** engaged with the housing **502** of the third embodiment of an EV lockout device **500** of FIGS. 13A-13C or the housing **1102** of the seventh embodiment of an EV lockout device **1100** of FIGS. 31A-31C, discussed below.

[0155] In some embodiments, the first elongated member **202** may be made of a flexible, but sturdy material, so that the hook portion **212** can slide up and over the locking mechanism **30** of the first EV charge port **20** (or the locking mechanism **30'** of the second EV charge port **20'** or the locking mechanism **54''** of the fourth EV charge plug **50''**) when the clip **200** is attached to or engaged with the housing **102** of the EV lockout device **100** (or the housing **502** of the EV lockout device **500** or the housing **1102** of the EV lockout device **1100**), but is not able to be easily tampered with or broken. In some embodiments, the first elongated member **202** also may be imparted with a length that is suitable for the hook portion **212** to slide up and over the locking mechanism **30** of the first EV charge port **20** (or the locking mechanism **30'** or the locking mechanism **54''**) and secure the EV lockout device **100** to the locking member **32** of the locking mechanism **30**, when the clip **200** is attached or engaged with the housing **102** of the EV lockout device **100** (or the housing **502** or the housing **1102**).

[0156] The second end **206** of the first elongated member **202** may be connected to or attached to a platform portion **220**. The platform portion **220** may include a first support leg **222a** disposed on a first platform side **224a** of the platform portion **220** and a second support leg **222b** disposed on a second platform side **224b** of the platform portion **220**. In some embodiments, the second support leg **222b** may be substantially parallel to and a mirror image of the first support leg **222a**. The first and second support legs **222a**, **222b** may extend from a front end **226** of the platform portion **220** on an angle downwardly and away from the front end **226**. In some embodiments, the angle at which the first and second support legs **222a**, **222b** may extend downwardly from the front end **226** of the platform portion **220** may be 45 degrees, or about 45 degrees. In other embodiments, the angle may be greater than or less than 45 degrees. In some embodiments, the first and second support legs **222a**, **222b** may be made of a flexible, but sturdy material, so that the first and second support legs **222a**, **222b** can compress and expand without breaking. In some embodiments, the first and second support legs **222a**, **222b** may be springs or a flexible material.

[0157] In some embodiments, the platform portion **220** may be imparted with a substantially rectangular shape. In other embodiments, the platform portion **220** may be imparted with another shape (e.g., elliptical) or an irregular shape. In some embodiments, the platform portion **220** may be imparted with a length and a width and/or a surface area that is large enough for a human finger or thumb to engage with and press downward on the platform portion **220**. For example, in some embodiments, the platform portion **220** may be imparted with a length of 14.88 mm, or about 14.88

mm, and a width of 22.5 mm, or about 22.5 mm.

[0158] The clip **200** may further include a second elongated member **228** that extends downwardly from a center portion **230** of the platform portion **220**. The second elongated member **228** may have a top end **232** and a bottom end **234** and may be connected or attached to the center portion **230** of the platform portion **220** at the top end **232** of the second elongated member **228**. In some embodiments, the second elongated member **228** may extend downwardly from the platform portion **220** such that the first elongated member **202** and second elongated member **228** are disposed substantially perpendicular to one another as shown in FIG. 3B. In some embodiments, the second elongated member **228** may be imparted with a length such that a space **235** exists between the bottom support **152** of the housing **102** of the EV lockout device **100** (or the bottom support **552** of the housing **502** of the EV lockout device **500** or the bottom support **1152** of the housing **1102** of the EV lockout device **1100**) and the bottom end **234** of the second elongated member **228** when the clip **200** is attached to or engaged with the housing **102** as shown in FIGS. 4B and 4C (or the housing **502** of FIGS. 13B and 3C or the housing **1102** of FIGS. 31B and 31C).

[0159] A lock through hole **236** may be disposed in and extend entirely through the second elongated member **228** between the top end **232** and the bottom end **234**. In some embodiments, the lock through hole **236** may be disposed adjacent the bottom end **234**. The lock through hole **236** may be imparted with a size and shape that is suitable for a shackle or shank of a lock (e.g., shackle **14** of lock **10**) to pass through. In some embodiments, the lock through hole **236** may be imparted with the same shape and/or size as the first and second holes **164a**, **164b** of the clip support unit **150** of the housing **102** (or the first and second holes **564a**, **564b** of the clip support unit **550** or the first and second holes **1164a**, **1164b** of the clip support unit **1150**). In some embodiments, the lock through hole **236** may be the same shape, but a different size as the first and second holes **164a**, **164b** of the clip support unit **150** of the housing **102**. In other embodiments, the lock through hole **236** may be a different size and/or shape as the first hole **164a** or the second hole **164b** of the clip support unit **150** of the housing **102** (or the first and second holes **564a**, **564b** of the clip support unit **550** or the first and second holes **1164a**, **1164b** of the clip support unit **1150**). In some embodiments, the lock through hole **236** may be disposed at a location in the second elongated member **228** such that the lock through hole **236** substantially aligns with the first hole **164a** and the second hole **164b** of the clip support unit **150** (or the first and second holes **564a**, **564b** of the clip support unit **550** or the first and second holes **1164a**, **1164b** of the clip support unit **1150**) when the clip **200** is attached to or engages with the housing **102** in a first clip position (see, e.g., FIG. 4A and 4C; or the housing **502** of FIGS. 13A and 13C or the housing **1102** of FIGS. 31A and 31C).

[0160] As shown in FIGS. 4A-4C, the clip **200** may attach to or otherwise engage with the housing **102**. When in the first clip position, the clip **200** may be placed on the housing **102** such that the coupling member **216**, the bottom side **210** of the first elongated member **202**, and/or the projection **218** of the clip **200** may engage with the clip connector **170** of the housing **102** as best seen in FIG. 4C. The clip connector **170** may rest within the nook **217** when in the first clip position, and the coupling member **216** may extend through the opening **174** of the housing **102** as shown in FIG. 4C. In addition, when in the first clip position, the first support leg **222a** may engage with the first clip support **166a** of the clip support unit **150**, and the second support leg **222b** may engage with the second clip support **166b** of the clip support unit **150**. The first elongated member **202** of the clip **200** may extend over the top flange portion **125** of the first flange **124**. The second elongated member **228** may be disposed within the gap **162** between the first side **154a** and the second side **154b** of the clip support unit **150** as shown in FIG. 4B. Further, when in the first clip position, the lock through hole **236** of the clip **200** may substantially align with the first hole **164a** of the first side **154a** and the second hole **164b** of the second side **154b** of the clip support unit **150**.

[0161] As discussed above, when in the first clip position, the first and second support legs **222a**, **222b** may engage with the first and second clip supports **166a**, **166b**, respectively. In addition, when in the first clip position, the lock through hole **236** of the second elongated member **228** of

the clip **200** may substantially align with the first hole **164a** of the first side **154a** and the second hole **164b** of the second side **154b** of the clip support unit **150** of the housing **102**.

[0162] When the EV lockout device **100** is being placed on or is being removed from the first EV charge port **20**, the clip **200** may move to a second clip position (not shown). When moving into the second clip position, the first end **204** of the first elongated member **202** may move in a substantially upward direction. In some embodiments, the clip **200** may move to the second clip position in response to the platform portion **220** being pressed downward so that the first and second support legs **222a**, **222b** compress (e.g., the angle between the platform portion **220** and one or more of the first and second support legs **222a**, **222b** decreases). In other embodiments, the clip **200** may move to the second clip position in response to the hook portion **212** of the first elongated member **202** engaging with the locking mechanism **30** of the first EV charge port **20**. In the second clip position, the first end **204** of the first elongated member **202** may raise up in order for the hook portion **212** to slide over the locking member **32** of the locking mechanism **30** of the first EV charge port **20**. As the first end **204** of the first elongated member **202** raises upward, the coupling member **216** and the projection **218** may move or rotate about the clip connector **170** of the housing **102** in a first direction. In some embodiments, as the first end **204** of the first elongated member **202** raises upward, the second elongated member **228** may move in a substantially downward direction such that the space **235** between the bottom end of the second elongated member **228** and the bottom support **152** of the clip support unit **150** decreases. In some embodiments, as the second elongated member **228** moves in the substantially downward direction, the lock through hole **236** of the second elongated member **228** may become out of alignment with the first hole **164a** and the second hole **164b** of the clip support unit **150**.

[0163] If the EV lockout device **100** is being placed onto the first EV charge port **20**, the clip **200** may move from the first clip position to the second clip position. As the clip **200** is moving from the first clip position to the second clip position, the housing **102** may engage with the first EV charge port **20**. In some embodiments, the first flange **124** of the housing **102** may engage with the first protrusion **34** of the first EV charge port **20** such that the first flange **124** surrounds all or a portion of the first protrusion **34**, the ridge **128** may engage with the channel **44** of the first EV charge port **20** such that the ridge **128** slides through the channel **44**, and the second flange **126** of the housing **102** may engage with the second protrusion **38** of the first EV charge port **20** such that the second flange **126** surrounds all or a portion of the second protrusion **38**.

[0164] After moving into the second clip position in response to the platform portion **220** being depressed, the clip **200** may move back into the first clip position upon release of the platform portion **220**. As the clip **200** moves back into the first clip position, the first end **204** of the first elongated member **202** may lower and the coupling member **216** and the projection **218** may move or rotate about the clip connector **170** of the housing **102** in a direction opposition the first direction. In addition, as the clip **200** moves back in the first clip position, the second elongated member **228** may move in a substantially upward direction as the first and second support legs **222a**, **222b** decompress (e.g., the angle between the platform portion **220** and one or more of the first and second support legs **222a**, **222b** increases). If the EV lockout device **100** has been placed on the first EV charge port **20**, the hook portion **212** may be disposed on the inner side **33** of the locking member **32** when the clip **200** returns to the first clip position. Additionally, when the clip **200** moves back into the first clip position, the first hole **164a** of the clip support unit **150**, the lock through hole **236** of the second elongated member **228**, and the second hole **164b** of the clip support unit **150** may be substantially aligned such that the shackle **14** of lock **10** may be threaded through the first hole **164a**, the lock through hole **236**, and the second hole **164b**, or vice versa. The shackle **14** may then be securely attached to the body **12** of the lock **10**, thereby placing the EV lockout device **100** in a locked position as shown in FIG. 5.

[0165] When in the locked position, the clip **200** cannot fully move to the second clip position (e.g., the hook portion **212** cannot be raised high enough to slide back over the locking member **32**

of the first EV charge port **20**) as the shackle **14** prevents the second elongated member **228** of the clip **200** from moving sufficiently downward. To remove the EV lockout device **100** from the first EV charge port **20**, the shackle **14** of the lock **10** may be disengaged from the body **12** of the lock **10** and removed from the clip support unit **150** of the housing **102** by threading the shackle **14** through the second hole **164b**, the lock through hole **236**, and the first hole **164a**, or vice versa. Once the lock **10** is removed from the EV lockout device **100**, the platform portion **220** of the clip **200** may be depressed causing the clip **200** to move from the first clip position to the second clip position in which the first and second support legs **222a**, **222b** compress and the hook portion **212** is raised. The hook portion **212** may then be moved over and away from the locking member **32** as the EV lockout device **100** is slid off or otherwise removed from the first EV charge port **20** (e.g., the first flange **124** is removed from the first protrusion **34**, the ridge **128** is removed from the channel **44**, and the second flange **126** is removed from the second protrusion **38**).

[0166] FIG. **6** depicts a partially exploded view of the lock **10**, a second embodiment of an EV lockout device **300**, and a first electric vehicle (EV) charge plug **50**. The first EV charge plug **50** may be attached to an EV charging station (not shown). In some embodiments, the first EV charge plug **50** may be provided in the form of a Combined Charging System (CCS) Type 1 plug.

[0167] The first EV charge plug **50** may include a plug housing **52** having a locking mechanism **54** disposed on a front side **55** thereof and protruding outwardly therefrom. The locking mechanism **54** may include a locking member **56** and, in some embodiments, the locking member **56** may be provided in the form of a hook. The first EV charge plug **50** may also include a lock release button **58** on a backside **60** of the plug housing **52**. In some embodiments, the lock release button **58** may be used to release or otherwise disengage the locking mechanism **54** and locking member **56** so that the EV lockout device **300** can be removed from the first EV charge plug **50**.

[0168] The first EV charge plug **50** may include a first protrusion **64** disposed on the front side **55** of the plug housing **52**. A first set of terminals **66** may be disposed within the first protrusion **64**. The first protrusion **64** may be imparted with a first protrusion shape. In some embodiments, the first protrusion shape may be substantially cylindrical with a substantially circular cross-section. In other embodiments, the first protrusion **64** may be imparted with other shapes, for example, a rectangular or square shape, a triangular shape, a trapezoidal shape, and the like, a combination of shapes, or an irregular shape. In some embodiments, the first protrusion **64** may be imparted with a slanted front side (not shown) where a top portion of the first protrusion has a first depth and the bottom portion of the first protrusion **64** has a second or max depth that is larger than the first depth (e.g., the depth of the protrusion is not uniform along the surface thereof).

[0169] The first protrusion **64** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the first protrusion **64** may include a diameter, a circumference, and/or a max depth. In some embodiments, the diameter of the first protrusion **64** may be about 44 mm. In some embodiments, the circumference of the first protrusion **64** may be about 138.23 mm. In some embodiments, the max depth of the first protrusion **64** may be about 33.5 mm.

[0170] The first EV charge plug **50** may include a second protrusion **68** disposed on the front side **55** of the plug housing **52**. A second set of terminals **70** may be disposed within the second protrusion **68**.

[0171] The second protrusion **68** may be imparted with a second protrusion shape. In some embodiments, the second protrusion shape may be substantially cylindrical with a substantially elliptical cross-section. In other embodiments, the second protrusion **68** may be imparted with other shapes, for example, a rectangular or square shape, a triangular shape, a trapezoidal shape, and the like, a combination of shapes, or an irregular shape,

[0172] The second protrusion **68** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the second protrusion **68** may include one or more diameters, a circumference, and/or a depth. One or more of the dimensions within the set of

dimensions of the second protrusion **68** may be the same or different than one or more of the dimensions within the set of dimensions for the first protrusion **64**. In one embodiment, the depth of the second protrusion **68** may be the same as the max depth of the first protrusion **64**.

[0173] In some embodiments, a first diameter (or width) of the second protrusion **68** may be about 59 mm, and a second diameter (or a height) of the second protrusion **68** may be about 32.6 mm. In some embodiments, the circumference of the second protrusion **68** may be about 155.22 mm. In some embodiments, the depth of the second protrusion **68** may be about 33.5 mm.

[0174] In some embodiments, the first protrusion **64** may be disposed above the second protrusion **68**. A protrusion ridge **62** may be disposed between the first protrusion **64** and the second protrusion **68**. The protrusion ridge **62** may connect the first protrusion **64** to the second protrusion **68**.

[0175] The protrusion ridge **62** may be imparted with a substantially rectangular shape, although other shapes may be used. The protrusion ridge **62** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the protrusion ridge **62** may include a width, a height, and a depth. For example, the width of the protrusion ridge **62** may be about 9 mm. In some embodiments, the height of the protrusion ridge **62** may be about 4.34 mm. In some embodiments, the depth of the protrusion ridge **62** may be about 33.5 mm.

[0176] The depth of the protrusion ridge **62** may be the same or different than the max depth of the first protrusion **64** or the depth of the second protrusion **68**. In some embodiments, the depth of the protrusion ridge **62**, the max depth of the first protrusion **64**, and the depth of the second protrusion **68** may be the same.

[0177] Turning to FIGS. 7A-7D, the second embodiment of the EV lockout device **300** may include a housing **302**. The housing **302** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the housing **302** may be made of ABS polymer plastic. In some embodiments, the housing **302** may be formed using injection molding or a 3D printer.

[0178] The housing **302** may include a base **304** defined by a front side **306** and a back side **308**. The base **304** may also have a top portion **310** and a bottom portion **312**, opposite the top portion **310**. The base **304** may have an outer edge **314** that defines a shape of the base **304**. In some embodiments, the shape of the base **304** may be substantially a square, a rectangle, a circle, an oval, a triangle, a trapezoid, and the like. In some embodiments, the shape of the base **304** may be imparted with round edges as shown in FIGS. 7A, 7B, and 7D. Combinations of shapes and irregular shapes are also contemplated.

[0179] In some embodiments, the base **304** may be imparted with a substantially rounded triangular shape as shown in FIGS. 7B and 7D where a top length **316** dimension of the top portion **310** is less than a bottom length **318** dimension of the bottom portion **312**. In some embodiments, the top length **316** of the top portion **310** may be 50.12 mm, or about 50.12 mm, and the bottom length **318** of the bottom portion **312** may be 67 mm, or about 67 mm.

[0180] The base **304** may be imparted with a base height **320**. In some embodiments, the base height **320** may be 86.27 mm, or about 86.27 mm. The base **304** also may be imparted with a base thickness (not shown). In some embodiments, the base thickness may be 3 mm, or about 3 mm.

[0181] The front side **306** of the base **304** may include a first flange **324** and a second flange **326** protruding outwardly therefrom. The first flange **324** may have a top flange portion **3325** and a bottom flange portion **327**. In some embodiments, the first flange **324** may be disposed above the second flange **326**. The first flange **324** may be connected to the second flange **326** by a flange channel **328**. In some embodiments, the flange channel may extend through a center area of the bottom flange portion **327**.

[0182] The first flange **324** may have a first flange wall **330** that is imparted with a first flange thickness. In some embodiments, the first flange thickness may be 2.4 mm, or about 2.4 mm.

[0183] The first flange wall **330** may define a first flange shape. The first flange shape may be any

size and shape suitable to engage with the first protrusion **64** when the EV lockout device **300** is placed on or otherwise in communication with the first EV charge plug **50**. In some embodiments, the first flange shape may be the same or substantially the same as the shape of the first protrusion **64** of the first EV charge plug **50**. In some embodiments, the first flange shape may be substantially cylindrical with a substantially circular cross-section.

[0184] The first flange **324** may be imparted with a first set of dimensions. In some embodiments, the first set of dimensions of the first flange **324** may be complimentary to the set of dimensions for the first protrusion **64** of the first EV charge plug **50** so that the first flange **324** can be placed onto and fit over the first protrusion **64**. In some instances, a friction fit between the first flange wall **330** and the first protrusion **64** may prevent the housing **302** from disengaging from the first EV charge plug **50**.

[0185] In some embodiments, the first set of dimensions of the first flange **324** may include a first flange diameter **332** (see FIG. 7B), a first flange circumference (not shown), and a first flange depth **334** (see FIG. 9C). In some embodiments, the first flange diameter **332** may be the same as or greater than the diameter of the first protrusion **64** of the first EV charge plug **50**. For example, in some embodiments, the first flange diameter **332**, may be 45.58 mm, or about 45.58 mm, and the first flange circumference may be 143.19 mm, or about 143.19 mm.

[0186] The first flange depth **334** may be the same as or different from the max depth of the first protrusion **64** of the first EV charge plug **50**. In some embodiments, the first flange depth **334** is the same as, substantially the same as, or greater than the max depth of the first protrusion **64** of the first EV charge plug **50**. For example, in some embodiments, the first flange depth **334** may be 34 mm, or about 34 mm.

[0187] The second flange **326** may be imparted with a second flange wall **336** that may be imparted with a thickness. In some embodiments, the second flange thickness may be 2.4 mm, or about 2.4 mm.

[0188] The second flange wall **336** may define a second flange shape. The second flange shape may be any size and shape suitable to engage with the second protrusion **68** when the EV lockout device **300** is placed on or otherwise in communication with the first EV charge plug **50**. In some embodiments, the second flange shape may be the same or substantially the same as the second protrusion **68** of the first EV charge plug **50**. In some embodiments, the second flange shape may be substantially cylindrical with a substantially elliptical cross-section.

[0189] The second flange **326** may be imparted with a second set of dimensions. In some embodiments, the second set of dimensions of the second flange **326** may be complimentary to the set of dimensions for the second protrusion **68** of the first EV charge plug **50** so that the second flange **326** can be placed onto and fit over the second protrusion **68**. In some instances, a friction fit between the second flange wall **336** and the second protrusion **68** may prevent the housing **302** from disengaging from the first EV charge plug **50**.

[0190] In some embodiments, the second set of dimensions of the second flange **326** may include a second flange diameter **338**, a second flange height **340** (see FIG. 7B), a second flange circumference (not shown), and a second flange depth **342** (see FIG. 9C). In some embodiments, the second flange diameter **338** may be the same as or greater than the first diameter of the second protrusion **68** of the first EV charge plug **50**. In some embodiments, the second flange height **340** may be the same as or greater than the second diameter of the second protrusion **68** of the first EV charge plug **50**. For example, in some embodiments, the second flange diameter **338** may be 61.58 mm, or about 61.58 mm, the second flange height **340** may be 34.58 mm, or about 34.58 mm, and the second flange circumference may be 162.65 mm, or about 162.65 mm.

[0191] The second flange depth **342** may be the same as or different from the depth of the second protrusion **68** of the first EV charge plug **50**. In some embodiments, the second flange depth **342** may be the same as, substantially the same as, or greater than the depth of the second protrusion **68** of the first EV charge plug **50**. In some embodiments, the second flange depth **342** may be the same

as the first flange depth **334**. For example, in some embodiments, the second flange depth **342** may be 34 mm, or about 34 mm.

[0192] The flange channel **328** may be imparted with a size and shape that is designed to engage with the protrusion ridge **62** of the first EV charge plug **50**. In some embodiments, the flange channel **328** may be imparted with a substantially rectilinear shape, although other shapes may be used that can engage with the protrusion ridge **62** of the first EV charge plug **50**.

[0193] The flange channel **328** may be imparted with a flange channel width **344**, a flange channel height **346** (see FIG. 7B), and a flange channel depth **348** (see FIG. 9C). The flange channel width **344** may be the same as or less than the width of the protrusion ridge **62** of the first EV charge plug **50**. In some embodiments, the flange channel width **344** may be 9 mm, or about 9 mm. The flange channel height **346** may be the same as or greater than the height of the protrusion ridge **62** of the first EV charge plug **50**. In some embodiments, the flange channel height **346** may be 2.53 mm, or about 2.53 mm.

[0194] The flange channel depth **348** may be the same as or different than the depth of the protrusion ridge **62** of the first EV charge plug **50**. In addition, the flange channel depth **348** may be the same as or different than the first flange depth **334** or the second flange depth **342**. In some embodiments, the flange channel depth **348**, the first flange depth **334**, and the second flange depth **342** may be the same. In some embodiments, the flange channel depth **348** may be 34 mm, or about 34 mm.

[0195] The back side **308** of the base **304** may include a clip support unit **350** protruding outwardly therefrom. The clip support unit **350** may be defined by a bottom support **352**, a first side **354a** and a second side **354b**. In some embodiments, the second side **354b** may be a mirror image of the first side **354a**. In some embodiments, the first side **354a** may be substantially parallel to the second side **354b**.

[0196] The first side **354a** may include a first side plate **356a** having a first bottom end **358a** and a first top end **360a**. The first side plate **356a** may be attached to the bottom support **352** on the first bottom end **358a**. The second side **354b** may include a second side plate **356b** having a second bottom end **358b** and a second top end **360b**. The second side plate **356b** may be attached to the bottom support **352** on the second bottom end **358b**.

[0197] The bottom support **352** may connect the first bottom end **358a** of the first side plate **356a** to the second bottom end **358b** of the second side plate **356b**. A gap **362** is formed between the first side plate **356a** and the second side plate **356b** as shown in FIG. 7D.

[0198] A first hole **364a** may be disposed in and extend entirely through the first side plate **356a** between the first top end **360a** and the first bottom end **358a**. A second hole **364b** may also be disposed in and extend entirely through the second side plate **356b** between the second top end **360b** and the second bottom end **358b**. The second hole **364b** may be disposed in the second side plate **356b** at the same location as the first hole **364a** is disposed in the first side plate **356a** so that the first hole **364a** and the second hole **364b** are substantially aligned. The first and second holes **364a**, **364b** may be any size and shape suitable for a shackle or shank of a lock (e.g., shackle **14** of the lock **10**) to pass through. In some embodiments, the first and second holes **364a**, **364b** may be imparted with a substantially oval shape. In other embodiments, the first and second holes **364a**, **364b** may be imparted with a substantially circular shape.

[0199] The first side **354a** may further include a first clip support **366a** on the first top end **360a**, and the second side **354b** may include a second clip support **366b** on the second top end **360b**. A first vertical support **368a** may extend from the first clip support **366a** upwards and past the top portion **310** of the base **304**. A second vertical support **368b** may extend from the second clip support **366b** upwards and past the top portion **310** of the base **304**. In some embodiments, the first vertical support **368a** and the second vertical support **368b** may be mirror images of each other.

[0200] A clip connector **370** may extend between a first top **372a** of the first vertical support **368a** and a first top **372b** of the second vertical support **368b** thereby connecting the first vertical support

368a to the second vertical support **368b**. The clip connector **370** may also extend from the first vertical support **368a** to the second vertical support **368b** in such a manner that an opening **374** is defined by the first vertical support **368a**, the second vertical support **368b**, the top flange portion **325** of the first flange **324**, and the clip connector **370**. In some embodiments, the clip connector **370** may be imparted with a cylindrical or rod shape.

[0201] A first front portion **378a** of the first vertical support **368a** may be connected or attached to the top flange portion **325** of the first flange **324**. A second front portion **378b** of the second vertical support **368a** may be connected or attached to the top flange portion **325** of the first flange **324**. Extending outward from the first front portion **378a** of the first vertical support **368a** may be a first guide rail **380a**, and extending outward from the second front portion **378b** of the second vertical support **368b** may be a second guide rail **380b**. In some embodiments, the first and second guide rails **380a**, **380b** may extend along the top flange portion **325** of the first flange **324** across the entire length of the top flange portion **325**. In some embodiments, the first and second guide rails **380a**, **380b** may be disposed substantially parallel to one another, and may be mirror images of each other.

[0202] The first guide rail **380a** may have a first center portion **382a**, and the second guide rail **380b** may have a second center portion **382b**. In some embodiments, a center divider **384** may extend from the first center portion **382a** of the first guide rail **380a** to the second center portion **382b** of the second guide rail **380b**, thereby connecting the first guide rail **380a** to the second guide rail **380b**. The center divider **384** may also be connected or attached to the top flange portion **325** of the first flange **324** on a bottom side of the center divider **384**. The center divider **384** may have an inner side **385** as shown in FIGS. 7A and 9C.

[0203] The clip support unit **350** may be designed to engage with one or more embodiments of a clip (e.g., **200**, **400**, **800**, **1000**) disclosed herein. In some embodiments, the first and second clip supports **366a**, **366b** may be designed to engage or otherwise be in communication with first and second support legs (e.g., **422a**, **422b**), respectively, of a clip (e.g., **400**). The clip connector **370** may be designed to engage with a coupling member (e.g., **416**) of the clip such that the coupling member can rotate about the clip connector **370** in order to raise and/or lower a first end (e.g., **404**) of a first elongated member (e.g., **402**) of the clip. In addition, the gap **362** disposed between the first and second side plates **356a**, **356b** of the clip support unit **350** may be designed to allow a second elongated member (e.g., **428**) of the clip to move in a substantially upwardly or downwardly direction between the first and second side plates **356a**, **356b**. The first and second guide rails **380a**, **380b** may be designed to engage or otherwise be in communication with first and second guide members (e.g., **412a**, **412b**), respectively, of the clip. Furthermore, the first and second holes **364a**, **364b** of the clip support unit **350** may be designed to substantially align with a lock through hole (e.g., **436**) of the clip to enable the shackle of a lock (e.g., shackle **14** of lock **10**) to pass through the housing **302** and clip and secure same to an EV charge port or an EV charge plug (e.g., **50**).

[0204] The second embodiment of the EV lockout device **300** may include a second embodiment of a clip **400** designed for use therewith. The clip **400** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the clip **400** may be made of polypropylene. In some embodiments, the clip **400** may be formed using injection molding or a 3D printer.

[0205] As shown in FIGS. 8A and 8B, the clip **400** may be provided in the form of a body having a first elongated member **402** defining a first end **404** and a second end **406**, opposite the first end **404**. The first elongated member **402** may have a first elongated side **407a** and a second elongated side **407b**, whereby the first and second elongated sides **407a**, **407b** may extend between the first end **404** and the second end **406** of the first elongated member **402**. The first elongated member **402** may further be defined by a top side **408** and a bottom side **410**.

[0206] The first elongated member **402** may further include a first guide member **412a** and a

second guide member **412b**. The first guide member **412a** may extend downwardly from the first elongated side **407a** and the second guide member **412b** may extend downwardly from the second elongated side **407b**. In some embodiments, the first and second guide members **412a**, **412b** may be imparted with a substantially square or rectangular shape. In other embodiments, the first and second guide members **412a**, **412b** may be imparted with a different shape.

[0207] The first end **404** may have a raised member **413**. The raised member **413** may project upwardly from the top side **408** of the first elongated member **402**. The raised member **413** may be designed to enable the first end **404** of the clip **400** to accommodate and slide over the locking mechanism **54** of the first EV charge plug **50** (or the locking mechanism **54'** of the second EV charge plug **50'** of FIG. 15, discussed below) as the clip **400** moves from a first clip position to a second clip position (as discussed in more detail below), and vice versa.

[0208] The first elongated member **402** may further include a coupling member **416** protruding downwardly from the bottom side **410** thereof. The coupling member **416** may be disposed on the bottom side **410** of the first elongated member **402** adjacent the second end **406**. In some embodiments, the bottom side **410** of the first elongated member **402** together with the coupling member **416** may define a “C” or “U” shape nook **417** as shown in FIG. 8B. The coupling member **416** may include a projection **418** on an end portion of the coupling member **416**. In some embodiments, the projection **418** may extend upwardly toward the bottom side **410** of the first elongated member **402**. In some embodiments, the nook **417** and the projection **418** may be designed to keep the clip **400** engaged with the housing **602** when placed thereon (see, e.g., FIG. 9C) or to keep the clip **400** engaged with the housing **602** of the fourth embodiment of an EV lockout device **600** of FIGS. 17A-17C.

[0209] In some embodiments, the first elongated member **402** may be made of a flexible, but sturdy material, so that the first end **404** can slide up and over the locking mechanism **54** of the first EV charge plug **50** (the locking mechanism **54'** of the second EV charge plug **50'**) when the clip **400** is attached to or engaged with the housing **302** of the EV lockout device **300** (or the housing **602** of the EV lockout device **600**)), but is not able to be easily tampered with or broken. In some embodiments, the first elongated member **402** may be imparted with a length that is suitable for the first elongated member **402** to engage with the locking mechanism **54** of the first EV charge plug **50** (the locking mechanism **54'**) and secure the EV lockout device **300** to the locking member **56**, when the clip **400** is attached to or engaged with the housing **302** of the EV lockout device **300** (or the housing **602** of the EV lockout device **600**).

[0210] The second end **406** of the first elongated member **402** may be connected to or attached to a platform portion **420**. The platform portion **420** may include a first support leg **422a** disposed on a first platform side **424a** of the platform portion **420** and a second support leg **422b** disposed on a second platform side **424b** of the platform portion **420**. In some embodiments, the second support leg **422b** may be substantially parallel to and a mirror image of the first support leg **422a**. The first and second support legs **422a**, **422b** may extend from a front end **426** of the platform portion **420** on an angle downward and away from the front end **426**. In some embodiments, the angle at which the first and second support legs **422a**, **422b** may extend downwardly from the front end **426** of the platform portion **420** may be 45 degrees or about 45 degrees. In other embodiments, the angle may be greater than or less than 45 degrees. In some embodiments, the first and second support legs **422a**, **422b** may be made of a flexible, but sturdy material, so that the first and second support legs **422a**, **422b** can compress and expand without breaking. In some embodiments, the first and second support legs **422a**, **422b** may be springs or a flexible material.

[0211] In some embodiments, the platform portion **420** may be imparted with a substantially rectangular shape. In other embodiments, the platform portion **420** may be imparted with another shape (e.g., elliptical) or an irregular shape. In some embodiments, the platform portion **420** may be imparted with a length and a width and/or a surface area that is large enough for a human finger or thumb to engage with and press downward on the platform portion **420**. For example, in some

embodiments, the platform portion **420** may be imparted with a length of 10.52 mm, or about 10.52 mm, and a width of 22 mm, or about 22 mm.

[0212] The clip **400** may further include a second elongated member **428** that extends downwardly from a center portion **430** of the platform portion **420**. The second elongated member **428** may have a top end **432** and a bottom end **434** and may be connected or attached to the center portion **430** of the platform portion **420** at the top end **432** of the second elongated member **428**. In some embodiments, the second elongated member **428** may extend downwardly from the platform portion **420** such that the first elongated member **402** and second elongated member **428** are disposed substantially perpendicular to one another as shown in FIG. 8B. In some embodiments, the second elongated member **428** may be imparted with a length such that a space **435** exists between the bottom support **352** of the housing **302** of the EV lockout device **300** (or the bottom support **652** of the housing **602** of the EV lockout device **600**) and the bottom end **434** of the second elongated member **428** when the clip **400** is attached to or engaged with the housing **302** of the EV lockout device **300** as shown in FIGS. 9B and 9C (or the housing **602** of FIGS. 17B and 17C).

[0213] A lock through hole **436** may be disposed in and extend entirely through the second elongated member **428** between the top end **432** and the bottom end **434**. In some embodiments, the lock through hole **436** may be disposed adjacent the bottom end **434**. The lock through hole **436** may be imparted with a size and shape that is suitable for a shackle or shank of a lock (e.g., shackle **14** of lock **10**) to pass through. In some embodiments, the lock through hole **436** may be imparted with the same shape and/or size as the first and second holes **364a**, **364b** of the clip support unit **350** of the housing **302** (or the first and second holes **664a**, **664b** of the clip support unit **650**). In some embodiments, the lock through hole **436** may be the same shape, but a different size as the first and second holes **364a**, **364b** of the clip support unit **350** of the housing **302**. In other embodiments, the lock through hole **436** may be a different size and/or shape as the first hole **364a** or the second hole **364b** of the clip support unit **350** of the housing **302** (or the first and second holes **664a**, **664b** of the clip support unit **650**). In some embodiments, the lock through hole **436** may be disposed at a location in the second elongated member **428** such that the lock through hole **436** substantially aligns with the first hole **364a** and the second hole **364b** of the clip support unit **350** (or the first and second holes **664a**, **664b** of the clip support unit **650**) when the clip **400** is attached to or engages with the housing **302** in a first clip position (see, e.g., FIGS. 9A and 9C; or the housing **602** of FIGS. 17A and 17C).

[0214] As shown in FIGS. 9A-9C, the clip **400** may attach to or otherwise engage with the housing **302**. When in the first clip position, the clip **400** may be placed on the housing **302** such that the coupling member **416**, the bottom side **410** of the first elongated member **402**, and/or the projection **418** of the clip **400** may engage with the clip connector **370** of the housing **302** as best seen in FIG. 9C. The clip connector **370** may rest within the nook **417** when in the first clip position, and the coupling member **416** may extend through the opening **374** of the housing **302** as shown in FIG. 4C. In addition, when in the first clip position, the first support leg **422a** may engage with the first clip support **366a** of the clip support unit **350**, and the second support leg **422b** may engage with the second clip support **366b** of the clip support unit **350**. The first elongated member **402** of the clip **400** may extend over the top flange portion **325** of the first flange **324**. The second elongated member **428** may be disposed within the gap **362** between the first side **354a** and the second side **354b** of the clip support unit **350** FIG. 9B. Further, when in the first clip position, the lock through hole **436** of the clip **400** may substantially align with the first hole **364a** of the first side **354a** and the second hole **364b** of the second side **354b** of the clip support unit **350**.

[0215] When in the first clip position, the first guide member **412a** of the clip **400** may be disposed on an outer side of the first guide rail **380a** on the top flange portion **376** of the first flange **324** of the housing **302**. In addition, the second guide member **412b** of the clip **400** may be disposed on an outer side of the second guide rail **380b** on the top flange portion **376** of the first flange **324** of the

housing **302**. In some embodiments, the first and second guide members **412a**, **412b** may be designed to prevent access to the locking mechanism **54** of the first EV charge plug **50** when in a locked position as shown in FIG. **10**.

[0216] As discussed above, when in the first clip position, the first and second support legs **422a**, **422b** may engage with the first and second clip supports **366a**, **366b**, respectively. In addition, when in the first clip position, the lock through hole **436** of the second elongated member **428** of the clip **400** may substantially align with the first hole **364a** of the first side **354a** and the second hole **364b** of the second side **354b** of the clip support unit **350** of the housing **302**. Further, when in the first clip position, the first and second guide members **412a**, **412b** may be disposed on the outer side of the first and second guide rails **380a**, **380b**, respectively.

[0217] When the EV lockout device **300** is being placed on or is being removed from the first EV charge plug **50**, the clip **400** may move to a second clip position (not shown). When moving into the second clip position, the first end **404** of the first elongated member **402** may move in a substantially upward direction. In some embodiments, the clip **400** may move to the second clip position either in response to the platform portion **420** being pressed downward so that the first and second support legs **422a**, **422b** compress (e.g., the angle between the platform portion **420** and one or more of the first and second support legs **422a**, **422b** decreases). In some embodiments, the clip **400** may move to the second clip position in response to the raised member **413** of the first end **404** of the first elongated member **402** engaging with the locking mechanism **54** of the first EV charge plug **50**. In other embodiments, the clip **400** may move to the second clip position in response to the locking member **56** of the first EV charge plug **50** engaging with the center divider **384** disposed between the first and second guard rails **380a**, **380b** on the top flange portion **376**. In the second clip position, the first end **404** of the first elongated member **402** may raise up in order for the locking member **56** to slide up and over the center divider **384** disposed between the first and second guard rails **380a**, **380b** on the top flange portion **376** of the housing **302**. As the first end **404** of the first elongated member **402** raises upward, the coupling member **416** and the projection **418** may move or rotate about the clip connector **370** of the housing **302** in a first direction. In some embodiments, as the first end **404** of the first elongated member **402** raises upward, the second elongated member **428** may move in a substantially downward direction such that the space **435** between the bottom end of the second elongated member **428** and the bottom support **352** of the clip support unit **350** decreases. In some embodiments, as the second elongated member **428** moves in the substantially downward direction, the lock through hole **436** of the second elongated member **428** may become out of alignment with the first hole **364a** and the second hole **364b** of the clip support unit **350**.

[0218] If the EV lockout device **300** is being placed onto the first EV charge plug **50**, the clip **400** may move from the first clip position to the second clip position. As the clip **400** is moving from the first clip position to the second clip position, the housing **302** may engage with the first EV charge plug **50**. In some embodiments, the first flange **324** of the housing **302** may engage with the first protrusion **64** of the first EV charge plug **50** such that the first flange **324** surrounds all or a portion of the first protrusion **64**, the flange channel **328** may engage with the protrusion ridge **62** of the first EV charge plug **50** such that the protrusion ridge **62** slides through the flange channel **328**, and the second flange **326** of the housing **302** may engage with the second protrusion **68** of the first EV charge plug **50** such that the second flange **326** surrounds all or a portion of the second protrusion **68**.

[0219] After moving into the second clip position in response to the platform portion **420** being depressed, the clip **400** may move back into the first clip position upon release of the platform portion **420**. As the clip **400** moves back into the first clip position, the first end **404** of the first elongated member **402** may lower and the coupling member **416** and the projection **418** may move or rotate about the clip connector **370** of the housing **302** in a direction opposition the first direction. In addition, as the clip **400** moves back in the first clip position, the second elongated

member **428** may move in a substantially upward direction as the first and second support legs **422a**, **422b** decompress (e.g., the angle between the platform portion **420** and one or more of the first and second support legs **422a**, **422b** increases). If the EV lockout device **300** has been placed on the first EV charge plug **50**, the locking member **56** may be disposed on the inner side **385** of the center divider **384**, when the clip **400** returns to the first clip position. Additionally, when the clip **400** moves back into the first clip position, the first hole **364a** of the clip support unit **350**, the lock through hole **436** of the second elongated member **428**, and the second hole **364b** of the clip support unit **350** may be substantially aligned such that the shackle **14** of lock **10** may be threaded through the first hole **364a**, the lock through hole **436**, and the second hole **364b**, or vice versa. The shackle **14** may then be securely attached to the body **12** of the lock **10**, thereby placing the EV lockout device **300** in a locked position as shown in FIG. **10**.

[0220] When in the locked position, the clip **400** cannot fully move to the second clip position (e.g., the first end **404** of the first elongated member **402** of the clip **400** cannot be raised high enough for the locking member **56** to slide up and back over the center divider **384** on the top flange portion **376** of the housing **302**) as the shackle **14** prevents the second elongated member **428** of the clip **400** from moving sufficiently downward. To remove the EV lockout device **300** from the first EV charge plug **50**, the shackle **14** of the lock **10** may be disengaged from the body **12** of the lock **10** and removed from the clip support unit **350** of the housing **302** by threading the shackle **14** through the second hole **364b**, the lock through hole **436**, and the first hole **364a**, or vice versa. Once the lock **10** is removed from the EV lockout device **300**, the platform portion **420** of the clip **400** may be depressed causing the clip **400** to move from the first clip position to the second clip position in which the first and second support legs **464a**, **464b** compress and the first end **404** of the first elongated member **402** is raised. The locking member **56** of the first EV charge plug **50** may then be moved up and over the center divider **384** as the EV lockout device **300** is slid off or otherwise removed from the first EV charge plug **50** (e.g., the first flange **324** is removed from the first protrusion **64**, the flange channel **328** is removed from the protrusion ridge **62**, and the second flange **326** is removed from the second protrusion **68**).

[0221] FIG. **11** depicts a partially exploded view of the lock **10**, a third embodiment of an EV lockout device **500**, and a second electric vehicle (EV) charge port **20''**.

[0222] The second EV charge port **20'** may be disposed inside of, on, or otherwise associated with an electric vehicle (not shown). In some embodiments, the second EV charge port **20'** may be provided in the form of a J1772 socket.

[0223] The second EV charge port **20'** may include a port housing **22'** having a first protrusion aperture **26'**. The first EV charge port **20'** may include a locking mechanism **30'** disposed above the first protrusion aperture **26'**. The locking mechanism **30'** may include a locking member **32'** having an inner side **33''**.

[0224] The second EV charge port **20'** may further include a first protrusion **34'** disposed within the first protrusion aperture **26'**. A first set of terminals receptors **36'** may be disposed within the first protrusion **34''**.

[0225] The first protrusion **34'** may be imparted with a first protrusion shape. In some embodiments, the first protrusion shape may be substantially cylindrical with a substantially circular cross-section. In other embodiments, the first protrusion **34'** may be imparted with other shapes, for example, a rectangular or square shape, a triangular shape, a trapezoidal shape, and the like, a combination of shapes, or an irregular shape.

[0226] The first protrusion **34'** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the first protrusion **34'** may include a diameter, a circumference, and/or a depth. In some embodiments, the diameter of the first protrusion **34'** may be about 33.53 mm. In some embodiments, the circumference of the first protrusion **34'** may be about 105.33 mm. In some embodiments, the depth of the first protrusion **34'** may be about 31.75 mm.

[0227] A channel **44'** may be disposed on a bottom area **45** of the first protrusion aperture **26'**. The channel **44'** may be defined by a first channel side **46a'**, a second channel side **46b**, and a channel bottom **48**. In some embodiments, the channel **44'** may be imparted with a substantially rectilinear shape, although other shapes may be used. The channel **44'** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the channel **44'** may include a width, a height, and a depth. For example, in some embodiments, the width of the channel **44'** may be about 8.56 mm. In some embodiments, the height of the channel **44'** may be about 4.3 mm. In some embodiments, the depth of the channel **44'** may be about 31.75 mm.

[0228] The depth of the channel **44** may be the same or different than the depth of the first protrusion **34'**. In some embodiments, the depths of the channel **44'** and the first protrusion **34'** may be the same.

[0229] Turning to FIGS. **12A-12D**, the third embodiment of the EV lockout device **500** may include a housing **502**. The housing **502** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the housing **502** may be made of ABS polymer plastic. In some embodiments, the housing **502** may be formed using injection molding or a 3D printer.

[0230] The housing **502** may include a base **504** defined by a front side **506** and a back side **508**. The base **504** may also have a top portion **510** and a bottom portion **512**, opposite the top portion **510**. In some embodiments, the top portion **510** terminates at a top edge **511**. The base **504** may have an outer edge **514** that defines a shape of the base **504**. In some embodiments, the shape of the base **504** may be substantially a square, a rectangle, a circle, an oval, a triangle, a trapezoid, and the like. In some embodiments, the shape of the base **504** may be imparted with round edges as shown in FIGS. **12A**, **12B**, and **12D**. Combinations of shapes and irregular shapes are also contemplated.

[0231] In some embodiments, the base **504** may be imparted with a combination shape as shown in FIGS. **12B** and **12D**, where the top portion **510** may be imparted with a substantially trapezoidal shape and the bottom portion **512** may have a substantially circular shape. The base **504** may be imparted with a top length **516** dimension of the top portion **510** and a bottom length **518** dimension of the bottom portion **512**. In some embodiments, the length **516** of the top portion **510** may be less than the length of the bottom portion **512**. In some embodiments, the length **516** of the top portion **510** may be 40.23 mm, or about 40.23 mm. In some embodiments, the length **518** of the bottom portion **512** may be 50 mm, or about 50 mm.

[0232] The base **504** may be imparted with a base height **520**. In some embodiments, the base height **520** may be 59 mm, or about 59 mm. The base **504** also may be imparted with a base thickness (not shown). In some embodiments, the base thickness may be 3 mm, or about 3 mm.

[0233] The top portion **510** of the base **504** may be interrupted by a divot **522** disposed in a center area of the top portion **510**. The divot **522** may extend downwardly from the top edge **511** and be defined by a first divot side **523a**, a second divot side **523b**, and a bottom divot side **523c**. In some embodiments, the divot **522** may be imparted with a "C" shape or a "U" shape as shown in FIGS. **12B** and **12D**.

[0234] The front side **506** of the base **504** may include a first flange **524** protruding outwardly therefrom. In some embodiments, the first flange **524** may be disposed below the divot **522** of the top portion **510**. The first flange **524** may have a top flange portion **525** and a bottom flange portion **527**. A ridge **528** may extend downwardly from a center area of the bottom flange portion **527**. The ridge **528** may extend along the entire length of the bottom flange portion **527** as shown in FIG. **12C**.

[0235] The first flange **524** may have a first flange wall **530** that is imparted with a first flange thickness. In some embodiments, the first flange thickness may be 2.5 mm, or about 2.5 mm.

[0236] The first flange wall **530** may define a first flange shape. The first flange shape may be any size and shape suitable to engage with the first protrusion **34'** when the EV lockout device **500** is placed on or otherwise in communication with the second EV charge port **20'**. In some

embodiments, the first flange shape may be the same or substantially the same as the shape of the first protrusion **34'** of the second EV charge port **20'**. In some embodiments, the first flange shape may be substantially cylindrical with a substantially circular cross-section.

[0237] The first flange **524** may be imparted with a first set of dimensions. In some embodiments, the first set of dimensions of the first flange **524** may be complimentary to the set of dimensions for the first protrusion **34'** of the second EV charge port **20'** so that the first flange **524** can be placed onto and fit over the first protrusion **34'**. In some instances, a friction fit between the first flange wall **530** and the first protrusion **34'** may prevent the housing **502** from disengaging from the second EV charge port **20''**.

[0238] In some embodiments, the first set of dimensions of the first flange **524** may include a first flange diameter **532** (see FIG. 12B), a first flange circumference (not shown), and a first flange depth **534** (see FIG. 13C). In some embodiments, the first flange diameter **532** may be the same as or greater than the diameter of the first protrusion **34'** of the second EV charge port **20'**. For example, in some embodiments, the first flange diameter **532**, may be 38 mm, or about 38 mm, and the first flange circumference may be 119.41 mm, or about 119.41 mm.

[0239] The first flange depth **534** may be the same as or different from the depth of the first protrusion **34'** of the second EV charge port **20'**. In some embodiments, the first flange depth **534** is the same as, substantially the same as, or less than the depth of the first protrusion **34'** of the second EV charge port **20'**. For example, in some embodiments, the first flange depth **534** may be 30 mm, or about 30 mm.

[0240] The ridge **528** may be imparted with a size and shape that is designed to engage with the channel **44''** of the second EV charge port **20'**. In some embodiments, the ridge **528** may be imparted with a substantially rectangular shape, although other shapes may be used that can engage with the channel **44'** of the second EV charge port **20''**.

[0241] The ridge **528** may be defined by a ridge width **544** (see FIG. 12B), a ridge height **546**, and a ridge depth **548** (see FIG. 12C). The ridge width **544** may be the same as or less than the width of the channel **44'** of the second EV charge port **20'**. In some embodiments, the ridge width **544** may be 6.49 mm, or about 6.49 mm. The ridge height **546** may be the same or greater than the height of the channel **44'** of the second EV charge port **20'**. In some embodiments, the ridge height **546** may be 1.54 mm, or about 1.54 mm.

[0242] The ridge depth **548** may be the same as or different than the depth of the channel **44'** of the second EV charge port **20'**. In addition, the ridge depth **548** may be the same as or different than the first flange depth **534**. In some embodiments, the ridge depth **548** and the first flange depth **534** may be the same. In some embodiments, the ridge depth **548** may be 30 mm, or about 30 mm.

[0243] The back side **508** of the base **504** may include a clip support unit **550** protruding outwardly therefrom. The clip support unit **550** may be defined by a bottom support **552**, a first side **554a** and a second side **554b**. In some embodiments, the first side **554a** may be substantially parallel to the second side **554b**. In some embodiments, the second side **554b** may be a mirror image of the first side **554a**.

[0244] The first side **554a** may include a first side plate **556a** having a first bottom end **558a** and a first top end **560a**. The first side plate **556a** may be attached to the bottom support **552** on the first bottom end **558a**. The second side **554b** may include a second side plate **556b** having a second bottom end **558b** and a second top end **560b**. The second side plate **556b** may be attached to the bottom support **552** on the second bottom end **558b**.

[0245] The bottom support **552** may connect the first bottom end **558a** of the first side plate **556a** to the second bottom end **558b** of the second side plate **556b**. A gap **562** is formed between the first side plate **556a** and the second side plate **556b** as shown in FIG. 12D.

[0246] A first hole **564a** may be disposed in and extend entirely through the first side plate **556a** between the first top end **560a** and the first bottom end **558a**. A second hole **564b** may also be disposed in and extend entirely through the second side plate **556b** between the second top end

560b and the second bottom end **558b**. The second hole **564b** may be disposed in the second side plate **556b** at the same location as the first hole **564a** is disposed in the first side plate **556a** so that the first hole **564a** and the second hole **564b** are substantially aligned. The first and second holes **564a**, **564b** may be any size and shape suitable for a shackle or shank of a lock (e.g., shackle **14** of the lock **10**) to pass through. In some embodiments, the first and second holes **564a**, **564b** may be imparted with a substantially oval shape. In other embodiments, the first and second holes **564a**, **564b** may be imparted with a substantially circular shape.

[0247] The first side **554a** may include a first clip support **566a** on the first top end **560a**, and the second side **554b** may include a second clip support **566b** on the second top end **560b**. A first vertical support **568a** may extend from the first clip support **566a** toward the top portion **510** of the housing **502** and a second vertical support **568b** may extend from the second clip support **566b** toward the top portion **510** of the housing **502**. In some embodiments, the first vertical support **568a** and the second vertical support **568b** may be mirror images of each other. The first and second vertical supports **568a**, **568b** may also be disposed on the back side **508** of the housing **502** adjacent the divot **522** such that the first vertical support **568a** is disposed adjacent the first divot side **523a** of the divot **522** and the second vertical support **568b** is disposed adjacent the second divot side **523b** of the divot **522**.

[0248] A clip connector **570** may extend between a first top **572a** of the first vertical support **568a** and a first top **572b** of the second vertical support **568b** thereby connecting the first vertical support **568a** to the second vertical support **568b**. The clip connector **570** may also extend across the divot **522** in such a manner that an opening **574** is defined by the first divot side **523a**, the second divot side **523b**, the bottom divot side **523c**, and the clip connector **570**. In some embodiments, the clip connector **570** may be imparted with a cylindrical or rod shape.

[0249] The clip support unit **550** may be designed to engage with one or more embodiments of a clip (e.g., **200**, **400**, **800**, **1000**) disclosed herein. In some embodiments, the first and second clip supports **566a**, **566b** may be designed to engage or otherwise be in communication with first and second support legs (e.g., **222a**, **222b**), respectively, of a clip (e.g., **200**). The clip connector **570** may be designed to engage with a coupling member (e.g., **216**) of the clip such that the coupling member can rotate about the clip connector **570** in order to raise and/or lower a first end (e.g., **204**) of a first elongated member (e.g., **202**) of the clip. In addition, the gap **562** disposed between the first and second side plates **556a**, **556b** of the clip support unit **550** may be designed to allow a second elongated member (e.g., **228**) of the clip to move in a substantially upwardly or downwardly direction between the first and second side plates **556a**, **556b**. Furthermore, the first and second holes **564a**, **564b** of the clip support unit **550** may be designed to substantially align with a lock through hole (e.g., **236**) of the clip to enable the shackle of a lock (e.g., shackle **14** of lock **10**) to pass through the housing **502** and clip and secure same to an EV charge port (e.g., **20'**) or an EV charge plug.

[0250] The third embodiment of the EV lockout device **500** may include the clip **200** designed for use therewith. As shown in FIGS. **13A-13C**, the clip **200** may attach to or otherwise engage with the housing **502**. When in the first clip position, the clip **200** may be placed on the housing **502** such that the coupling member **216**, the bottom side **210** of the first elongated member **202**, and/or the projection **218** of the clip **200** may engage with the clip connector **570** of the housing **502** as best seen in FIG. **13C**. The clip connector **570** may rest within the nook **217** when in the first clip position and the coupling member **216** may extend through the opening **574** of the housing **502** as shown in FIG. **13C**. In addition, when in the first clip position, the first support leg **222a** may engage with the first clip support **566a** of the clip support unit **550**, and the second support leg **222b** may engage with the second clip support **566b** of the clip support unit **550**. The first elongated member **202** of the clip **200** may extend over the top flange portion **525** of the first flange **524**. The second elongated member **228** may be disposed within the gap **562** between the first side **554a** and the second side **554b** of the clip support unit **550** as shown in FIG. **13B**. Further,

when in the first clip position, the lock through hole **236** of the clip **200** may substantially align with the first hole **564a** of the first side **554a** and the second hole **564b** of the second side **554b** of the clip support unit **550**.

[0251] As discussed above, when in the first clip position, the first and second support legs **222a**, **222b** may engage with the first and second clip supports **566a**, **566b**, respectively. In addition, when in the first clip position, the lock through hole **236** of the second elongated member **228** of the clip **200** may substantially align with the first hole **564a** of the first side **554a** and the second hole **564b** of the second side **554b** of the clip support unit **550** of the housing **502**.

[0252] When the EV lockout device **500** is being placed on or is being removed from the second EV charge port **20''**, the clip **200** may move to a second clip position (not shown). When moving into the second clip position, the first end **204** of the first elongated member **202** may move in a substantially upward direction. In some embodiments, the clip **200** may move to the second clip position in response to the platform portion **220** being pressed downward so that the first and second support legs **222a**, **222b** compress (e.g., the angle between the platform portion **220** and one or more of the first and second support legs **222a**, **222b** decreases). In other embodiments, the clip **200** may move to the second clip position in response to the hook portion **212** of the first elongated member **202** engaging with the locking mechanism **30'** of the second EV charge port **20'**. In the second clip position, the first end **204** of the first elongated member **202** may raise up in order for the hook portion **212** to slide over the locking member **32'** of the locking mechanism **30'** of the second EV charge port **20'**. As the first end **204** of the first elongated member **202** raises upward, the coupling member **216** and the projection **218** may move or rotate about the clip connector **570** of the housing **502** in a first direction. In some embodiments, as the first end **204** of the first elongated member **202** raises upward, the second elongated member **228** may move in a substantially downward direction such that the space **235** between the bottom end of the second elongated member **228** and the bottom support **552** of the clip support unit **550** decreases. In some embodiments, as the second elongated member **228** moves in the substantially downward direction, the lock through hole **236** of the second elongated member **228** may become out of alignment with the first hole **564a** and the second hole **564b** of the clip support unit **550**.

[0253] If the EV lockout device **500** is being placed onto the second EV charge port **20'**, the clip **200** may move from the first clip position to the second clip position. As the clip **200** is moving from the first clip position to the second clip position, the housing **502** may engage with the second EV charge port **20'**. In some embodiments, the first flange **524** of the housing **502** may engage with the first protrusion **34'** of the second EV charge port **20'** such that the first flange **524** surrounds all or a portion of the first protrusion **34'** and the ridge **528** may engage with the channel **44'** of the second EV charge port **20'** such that the ridge **528** slides through the channel **44''**.

[0254] After moving into the second clip position in response to the platform portion **220** being depressed, the clip **200** may move back into the first clip position upon release of the platform portion **220**. As the clip **200** moves back into the first clip position, the first end **204** of the first elongated member **202** may lower and the coupling member **216** and the projection **218** may move or rotate about the clip connector **570** of the housing **502** in a direction opposite the first direction. In addition, as the clip **200** moves back in the first clip position, the second elongated member **228** may move in a substantially upward direction as the first and second support legs **222a**, **222b** decompress (e.g., the angle between the platform portion **220** and one or more of the first and second support legs **222a**, **222b** increases). If the EV lockout device **500** has been placed on the second EV charge port **20'**, the hook portion **212** may be disposed on the inner side **33'** of the locking member **32'**, when the clip **200** returns to the first clip position. Additionally, when the clip **200** moves back into the first clip position, the first hole **564a** of the clip support unit **550**, the lock through hole **236** of the second elongated member **228**, and the second hole **564b** of the clip support unit **550** may be substantially aligned such that the shackle **14** of lock **10** may be threaded through the first hole **564a**, the lock through hole **236**, and the second hole **564b**, or vice versa. The

shackle **14** may then be securely attached to the body **12** of the lock **10**, thereby placing the EV lockout device **500** in a locked position as shown in FIG. **14**.

[0255] When in the locked position, the clip **200** cannot fully move to the second clip position (e.g., the hook portion **212** cannot be raised high enough to slide back over the locking member **32'** of the second EV charge port **20'**) as the shackle **14** prevents the second elongated member **228** of the clip **200** from moving sufficiently downward. To remove the EV lockout device **500** from the second EV charge port **20'**, the shackle **14** of the lock **10** may be disengaged from the body **12** of the lock **10** and removed from the clip support unit **550** of the housing **502** by threading the shackle **14** through the second hole **564b**, the lock through hole **236**, and the first hole **564a**, or vice versa. Once the lock **10** is removed from the EV lockout device **500**, the platform portion **220** of the clip **200** may be depressed causing the clip **200** to move from the first clip position to the second clip position in which the first and second support legs **222a**, **222b** compress and the hook portion **212** is raised. The hook portion **212** may then be moved over and away from the locking member **32'** as the EV lockout device **500** is slid off or otherwise removed from the second EV charge port **20'** (e.g., the first flange **524** is removed from the first protrusion **34'** and the ridge **528** is removed from the channel **44**).

[0256] FIG. **15** depicts a partially exploded view of the lock **10**, a fourth embodiment of an EV lockout device **500**, and a second electric vehicle (EV) charge plug **50'**.

[0257] The second EV charge plug **50'** may be attached to an EV charging station (not shown). In some embodiments, the second EV charge plug **50'** may be provided in the form of a J1772 plug.

[0258] The second EV charge plug **50'** may include a plug housing **52'** having a locking mechanism **54'** disposed on a front side **55'** thereof and protruding outwardly therefrom. The locking mechanism **54'** may include a locking member **56'**, and, in some embodiments, the locking member **56'** may be provided in the form of a hook. The second EV charge plug **50'** may also include a lock release button **58'** on a backside **60'** of the plug housing **52'**. In some embodiments, the lock release button **58'** may be used to release or otherwise disengage the locking mechanism **54'** and locking member **56'** so that the EV lockout device **500** can be removed from the second EV charge plug **50'**.

[0259] The second EV charge plug **50'** may include a first protrusion **64'** disposed on the front side **55'** of the plug housing **52'**. A first set of terminals **66'** may be disposed within the first protrusion **64'**.

[0260] The first protrusion **64'** may be imparted with a first protrusion shape. In some embodiments, the first protrusion shape may be substantially cylindrical with a substantially circular cross-section. In other embodiments, the first protrusion **64'** may be imparted with other shapes, for example, a rectangular or square shape, a triangular shape, a trapezoidal shape, and the like, a combination of shapes, or an irregular shape. In some embodiments, the first protrusion **64'** may be imparted with a slanted front side (not shown) where a top portion of the first protrusion **64'** has a first depth and the bottom portion of the first protrusion **64'** has a second or max depth that is larger than the first depth (e.g., the depth of the protrusion is not uniform along the surface thereof).

[0261] The first protrusion **64'** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the first protrusion **64'** may include a diameter, a circumference, and/or a max depth. In some embodiments, the diameter of the first protrusion **64'** may be about 43.8 mm. In some embodiments, the circumference of the first protrusion **64'** may be about 137.6 mm. In some embodiments, the max depth of the first protrusion **64'** may be about 33.58 mm.

[0262] A protrusion ridge **62'** may be disposed below the first protrusion **64'**. The protrusion ridge **62'** may be imparted with a substantially rectangular shape, although other shapes may be used. The protrusion ridge **62'** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the protrusion ridge **62'** may include a width, a height, and a depth. For example, the width of the protrusion ridge **62'** may be about 8 mm. In some

embodiments, the height of the protrusion ridge **62'** may be about 2.69 mm. In some embodiments, the depth of the protrusion ridge **62'** may be about 33.58 mm.

[0263] The depth of the protrusion ridge **62'** may be the same or different than the max depth of the first protrusion **64'**. In some embodiments, the depth of the protrusion ridge **62'** and the max depth of the first protrusion **64** may be the same.

[0264] Turning to FIGS. **16A-16D**, the fourth embodiment of the EV lockout device **600** may include a housing **602**. The housing **602** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the housing **602** may be made of ABS polymer plastic. In some embodiments, the housing **602** may be formed using injection molding or a 3D printer.

[0265] The housing **602** may include a base **604** defined by a front side **606** and a back side **608**. The base **604** may also have a top portion **610** and a bottom portion **612**, opposite the top portion **610**. The base **604** may have an outer edge **614** that defines a shape of the base **604**. In some embodiments, the shape of the base **604** may be substantially a square, a rectangle, a circle, an oval, a triangle, a trapezoid, and the like. In some embodiments, the shape of the base **604** may be imparted with round edges as shown in FIGS. **16A**, **16B**, and **16D**. Combinations of shapes and irregular shapes are also contemplated.

[0266] In some embodiments, the base **604** may be imparted with a substantially circular shape as shown in FIGS. **16B** and **16D**. In some embodiments, the base **604** may be imparted with a base diameter (or width) **616**. In some embodiments, the base diameter (or width) **616** may be 51 mm, or about 51 mm.

[0267] The base **604** may be imparted with a base height **620**. In some embodiments, the base height **620** may be 51 mm, or about 51 mm. The base **604** also may be imparted with a base thickness (not shown). In some embodiments, the base thickness may be 3 mm, or about 3 mm.

[0268] The front side **606** of the base **604** may include a first flange **624** protruding outwardly therefrom. The first flange **624** may have a top flange portion **625** and a bottom flange portion **627**. A flange channel **628** may be disposed in a center area of the bottom flange portion **627**. In some embodiments, the flange channel **628** may extend along the entire length of the bottom flange portion **627**.

[0269] The first flange **624** may have a first flange wall **630** that is imparted with a first flange thickness. In some embodiments, the first flange thickness may be 2.4 mm, or about 2.4 mm.

[0270] The first flange wall **630** may define a first flange shape. The first flange shape may be any size and shape suitable to engage with the first protrusion **64'** when the EV lockout device **600** is placed on or otherwise in communication with the second EV charge plug **50'**. In some embodiments, the first flange shape may be the same or substantially the same as the shape of the first protrusion **64'** of the second EV charge plug **50'**. In some embodiments, the first flange shape may be substantially cylindrical with a substantially circular cross-section.

[0271] The first flange **624** may be imparted with a first set of dimensions. In some embodiments, the first set of dimensions of the first flange **624** may be complimentary to the set of dimensions for the first protrusion **64'** of the second EV charge plug **50'** so that the first flange **624** can be placed onto and fit over the first protrusion **64'**. In some instances, a friction fit between the first flange wall **630** and the first protrusion **64'** may prevent the housing **602** from disengaging from the second EV charge plug **50'**.

[0272] In some embodiments, the first set of dimensions of the first flange **624** may include a first flange diameter **632** (see FIG. **16B**), a first flange circumference (not shown), and a first flange depth **634** (see FIG. **17C**). In some embodiments, the first flange diameter **632** may be the same as or greater than the diameter of the first protrusion **64'** of the second EV charge port **50'**. For example, in some embodiments, the first flange diameter **632**, may be 45 mm, or about 45 mm, and the first flange circumference may be 141.37 mm, or about 141.37 mm.

[0273] The first flange depth **634** may be the same as or different from the max depth of the first

protrusion **64'** of the second EV charge plug **50'**. In some embodiments, the first flange depth **634** is the same as, substantially the same as, or greater than the max depth of the first protrusion **64'** of the second EV charge plug **50'**. For example, in some embodiments, the first flange depth **634** may be 34 mm, or about 34 mm.

[0274] The flange channel **628** may be imparted with a size and shape that is designed to engage with the protrusion ridge **62'** of the second EV charge plug **50'**. In some embodiments, the flange channel **628** may be imparted with a substantially rectilinear shape, although other shapes may be used that can engage with the protrusion ridge **62'** of the second EV charge plug **50'**.

[0275] The flange channel **628** may be imparted with a flange channel width **644** (see FIG. 16B), a flange channel height **646**, and a flange channel depth **648** (see FIG. 17C). The flange channel width **644** may be the same as or greater than the width of the protrusion ridge **62'** of the second EV charge plug **50'**. In some embodiments, the flange channel width **644** may be 9.58 mm, or about 9.58 mm. The flange channel height **646** may be the same or greater than the height of the protrusion ridge **62'** of the second EV charge plug **50'**. In some embodiments, the flange channel height **646** may be 2.4 mm, or about 2.4 mm.

[0276] The flange channel depth **648** may be the same as or different than the depth of the protrusion ridge **62'** of the second EV charge plug **50'**. In addition, the flange channel depth **648** may be the same as or different than the first flange depth **634**. In some embodiments, the flange channel depth **648** and the first flange depth **634** may be the same. In some embodiments, the flange channel depth **648** may be 34 mm, or about 34 mm.

[0277] The back side **608** of the base **604** may include a clip support unit **650** protruding outwardly therefrom. The clip support unit **650** may be defined by a bottom support **652**, a first side **654a**, and a second side **654b**. In some embodiments, the first side **654a** may be substantially parallel to the second side **654b**. In some embodiments, the second side **654b** may be a mirror image of the first side **654a**.

[0278] The first side **654a** may include a first side plate **656a** having a first bottom end **658a** and a first top end **660a**. The first side plate **656a** may be attached to the bottom support **652** on the first bottom end **658a**. The second side **654b** may include a second side plate **656b** having a second bottom end **658b** and a second top end **660b**. The second side plate **656b** may be attached to the bottom support **652** on the second bottom end **658b**.

[0279] The bottom support **652** may connect the first bottom end **658a** of the first side plate **656a** to the second bottom end **658b** of the second side plate **656b**. A gap **662** is formed between the first side plate **656a** and the second side plate **656b** as shown in FIG. 16D.

[0280] A first hole **664a** may be disposed in and extend entirely through the first side plate **656a** between the first top end **660a** and the first bottom end **658a**. A second hole **664b** may also be disposed in and extend entirely through the second side plate **656b** between the second top end **660b** and the second bottom end **658b**. The second hole **664b** may be disposed in the second side plate **656b** at the same location as the first hole **664a** is disposed in the first side plate **656a** so that the first hole **664a** and the second hole **664b** are substantially aligned. The first and second holes **664a**, **664b** may be any size and shape suitable for a shackle or shank of a lock (e.g., shackle **14** of the lock **10**) to pass through. In some embodiments, the first and second holes **664a**, **664b** may be imparted with a substantially oval shape. In other embodiments, the first and second holes **664a**, **664b** may be imparted with a substantially circular shape.

[0281] The first side **654a** may further include a first clip support **666a** on the first top end **660a**, and the second side **654b** may include a second clip support **666b** on the second top end **660b**. A first vertical support **668a** may extend from the first clip support **666a** upward and past the top portion **610** of the base **604**. A second vertical support **668b** may extend from the second clip support **666b** upward and past the top portion **610** of the base **604**. In some embodiments, the first vertical support **668a** and the second vertical support **668b** may be mirror images of each other.

[0282] A clip connector **670** may extend between a first top **672a** of the first vertical support **668a**

and a first top **672b** of the second vertical support **668b** thereby connecting the first vertical support **368a** to the second vertical support **668b**. The clip connector **670** may also extend from the first vertical support **668a** to the second vertical support **668b** in such a manner that an opening **674** is defined by the first vertical support **668a**, the second vertical support **668b**, the top flange portion **625** of the first flange **624**, and the clip connector **670**. In some embodiments, the clip connector **670** may be imparted with a cylindrical or rod shape.

[0283] A first front portion **678a** of the first vertical support **668a** may be connected or attached to the top flange portion **625** of the first flange **624**. A second front portion **678b** of the second vertical support **668a** may be connected or attached to the top flange portion **625** of the first flange **624**. Extending outward from the first front portion **678a** of the first vertical support **668a** may be a first guide rail **680a**, and extending outward from the second front portion **678b** of the second vertical support **668b** may be a second guide rail **680b**. In some embodiments, the first and second guide rails **680a**, **680b** may extend along the top flange portion **625** of the first flange **624** across the entire length of the top flange portion **325**. In some embodiments, the first and second guide rails **680a**, **680b** may be disposed substantially parallel to one another, and may be mirror images of each other.

[0284] The first guide rail **680a** may have a first center portion **682a**, and the second guide rail **680b** may have a second center portion **682b**. In some embodiments, a center divider **684** may extend from the first center portion **682a** of the first guide rail **680a** to the second center portion **682b** of the second guide rail **680b**, thereby connecting the first guide rail **680a** to the second guide rail **680b**. The center divider **684** may also be connected or attached to the top flange portion **625** of the first flange **624** on a bottom side of the center divider **684**. The center divider **684** may have an inner side **685** as shown in FIGS. **16A** and **17C**.

[0285] The clip support unit **650** may be designed to engage with one or more embodiments of a clip (e.g., **200**, **400**, **800**, **1000**) disclosed herein. In some embodiments, the first and second clip supports **666a**, **666b** may be designed to engage or otherwise be in communication with first and second support legs (e.g., **422a**, **422b**), respectively, of a clip (e.g., **400**). The clip connector **670** may be designed to engage with a coupling member (e.g., **416**) of the clip such that the coupling member can rotate about the clip connector **670** in order to raise and/or lower a first end (e.g., **404**) of a first elongated member (e.g., **402**) of the clip. In addition, the gap **662** disposed between the first and second side plates **656a**, **656b** of the clip support unit **650** may be designed to allow a second elongated member (e.g., **428**) of the clip to move in a substantially upwardly or downwardly direction between the first and second side plates **656a**, **656b**. The first and second guide rails **680a**, **680b** may be designed to engage or otherwise be in communication with first and second guide members (e.g., **412a**, **412b**), respectively, of the clip. Furthermore, the first and second holes **664a**, **664b** of the clip support unit **650** may be designed to substantially align with a lock through hole (e.g., **436**) of the clip to enable the shackle of a lock (e.g., shackle **14** of lock **10**) to pass through the housing **602** and clip and secure same to an EV charge port or an EV charge plug (e.g., **50**).

[0286] The fourth embodiment of the EV lockout device **600** may include the clip **400** designed for use therewith. As shown in FIGS. **17A-17C**, the clip **400** may attach to or otherwise engage with the housing **602**. When in the first clip position, the clip **400** may be placed on the housing **602** such that the coupling member **416**, the bottom side **410** of the first elongated member **402**, and/or the projection **418** of the clip **400** may engage with the clip connector **670** of the housing **602** as best seen in FIG. **17C**. The clip connector **670** may rest within the nook **417** when in the first clip position, and the coupling member **416** may extend through the opening **674** of the housing **302** as shown in FIG. **17C**. In addition, when in the first clip position, the first support leg **422a** may engage with the first clip support **666a** of the clip support unit **650**, and the second support leg **422b** may engage with the second clip support **666b** of the clip support unit **650**. The first elongated member **402** of the clip **400** may extend over the top flange portion **625** of the first

flange **624**. The second elongated member **428** may be disposed within the gap **662** between the first side **654a** and the second side **654b** of the clip support unit **650** as shown in FIG. 17B. Further, when in the first clip position, the lock through hole **436** of the clip **400** may substantially align with the first hole **664a** of the first side **654a** and the second hole **664b** of the second side **354b** of the clip support unit **650**.

[0287] When in the first clip position, the first guide member **412a** of the clip **400** may be disposed on an outer side of the first guide rail **680a** on the top flange portion **625** of the first flange **624** of the housing **602**. In addition, the second guide member **412b** of the clip **400** may be disposed on an outer side of the second guide rail **680b** on the top flange portion **625** of the first flange **624** of the housing **602**. In some embodiments, the first and second guide members **412a**, **412b** may be designed to prevent access to the locking mechanism **54'** of the second EV charge plug **50'** when in a locked position as shown in FIG. 18.

[0288] As discussed above, when in the first clip position, the first and second support legs **422a**, **422b** may engage with the first and second clip supports **666a**, **666b**, respectively. In addition, when in the first clip position, the lock through hole **436** of the second elongated member **428** of the clip **400** may substantially align with the first hole **664a** of the first side **654a** and the second hole **664b** of the second side **654b** of the clip support unit **350** of the housing **602**. Further, when in the first clip position, the first and second guide members **412a**, **412b** may be disposed on the outer side of the first and second guide rails **680a**, **680b**, respectively.

[0289] When the EV lockout device **400** is being placed on or is being removed from the second EV charge plug **50'**, the clip **400** may move to a second clip position (not shown). When moving into the second clip position, the first end **404** of the first elongated member **402** may move in a substantially upward direction. In some embodiments, the clip **400** may move to the second clip position either in response to the platform portion **420** being pressed downward so that the first and second support legs **422a**, **422b** compress (e.g., the angle between the platform portion **420** and one or more of the first and second support legs **422a**, **422b** decreases). In some embodiments, the clip **400** may move to the second clip position in response to the raised member **413** of the first end **404** of the first elongated member **402** engaging with the locking mechanism **54'** of the second EV charge plug **50'**. In other embodiments, the clip **400** may move to the second clip position in response to the locking member **56'** of the second EV charge plug **50'** engaging with the center divider **684** disposed between the first and second guard rails **680a**, **680b** on the top flange portion **625**. In the second clip position, the first end **404** of the first elongated member **402** may raise up in order for the locking member **56'** to slide up and over the center divider **684** disposed between the first and second guard rails **680a**, **680b** on the top flange portion **625** of the housing **602**. As the first end **404** of the first elongated member **402** raises upward, the coupling member **416** and the projection **418** may move or rotate about the clip connector **670** of the housing **602** in a first direction. In some embodiments, as the first end **404** of the first elongated member **402** raises upward, the second elongated member **428** may move in a substantially downward direction such that the space **435** between the bottom end of the second elongated member **428** and the bottom support **652** of the clip support unit **650** decreases. In some embodiments, as the second elongated member **428** moves in the substantially downward direction, the lock through hole **436** of the second elongated member **428** may become out of alignment with the first hole **664a** and the second hole **664b** of the clip support unit **650**.

[0290] If the EV lockout device **600** is being placed onto the second EV charge plug **50'**, the clip **400** may move from the first clip position to the second clip position. As the clip **400** is moving from the first clip position to the second clip position, the housing **602** may engage with the second EV charge plug **50'**. In some embodiments, the first flange **624** of the housing **602** may engage with the first protrusion **64'** of the second EV charge plug **50'** such that the first flange **624** surrounds all or a portion of the first protrusion **64'** and the flange channel **628** may engage with the protrusion ridge **62'** of the second EV charge plug **50'** such that the protrusion ridge **62'** slides through the

flange channel **628**.

[0291] After moving into the second clip position in response to the platform portion **420** being depressed, the clip **400** may move back into the first clip position upon release of the platform portion **420**. As the clip **400** moves back into the first clip position, the first end **404** of the first elongated member **402** may lower and the coupling member **416** and the projection **418** may move or rotate about the clip connector **670** of the housing **602** in a direction opposite the first direction. In addition, as the clip **400** moves back in the first clip position, the second elongated member **428** may move in a substantially upward direction as the first and second support legs **422a**, **422b** decompress (e.g., the angle between the platform portion **420** and one or more of the first and second support legs **422a**, **422b** increases). If the EV lockout device **600** has been placed on the second EV charge plug **50'**, the locking member **56'** may be disposed on the inner side **685** of the center divider **684**, when the clip **400** returns to the first clip position. Additionally, when the clip **400** moves back into the first clip position, the first hole **664a** of the clip support unit **650**, the lock through hole **436** of the second elongated member **428**, and the second hole **664b** of the clip support unit **650** may be substantially aligned such that the shackle **14** of lock **10** may be threaded through the first hole **664a**, the lock through hole **436**, and the second hole **664b**, or vice versa. The shackle **14** may then be securely attached to the body **12** of the lock **10**, thereby placing the EV lockout device **600** in a locked position as shown in FIG. **18**.

[0292] When in the locked position, the clip **400** cannot fully move to the second clip position (e.g., the first end **404** of the first elongated member **402** of the clip **400** cannot be raised high enough for the locking member **56'** to slide up and back over the center divider **684** on the top flange portion **625** of the housing **602**) as the shackle **14** prevents the second elongated member **428** of the clip **400** from moving sufficiently downward. To remove the EV lockout device **600** from the second EV charge plug **50'**, the shackle **14** of the lock **10** may be disengaged from the body **12** of the lock **10** and removed from the clip support unit **650** of the housing **602** by threading the shackle **14** through the second hole **664b**, the lock through hole **436**, and the first hole **664a**, or vice versa. Once the lock **10** is removed from the EV lockout device **600**, the platform portion **420** of the clip **400** may be depressed causing the clip **400** to move from the first clip position to the second clip position in which the first and second support legs **464a**, **464b** compress and the first end **404** of the first elongated member **402** is raised. The locking member **56'** of the second EV charge plug **50'** may then be moved up and over the center divider **684** as the EV lockout device **600** is slid off or otherwise removed from the second EV charge plug **50'** (e.g., the first flange **624** is removed from the first protrusion **64'** and the flange channel **628** is removed from the protrusion ridge **62''**).

[0293] FIG. **19** depicts a partially exploded view of the lock **10**, a fifth embodiment of an EV lockout device **700**, and a third electric vehicle (EV) charge plug **50''**.

[0294] The third EV charge plug **50''** may be attached to an EV charging station (not shown). In some embodiments, the third EV charge plug **50''** may be provided in the form of a Combined Charging System (CCS) Type 2 charge plug.

[0295] The third EV charge plug **50''** may include a plug housing **52''** having a front side **55''**. A first protrusion **64''** may extend from the front side **55''** of the plug housing **52''**. The first protrusion **64''** may have an inner wall **63**.

[0296] The third EV charge plug **50''** may include a locking mechanism **54''** disposed on a top side **65** of the first protrusion **64''**. In some embodiments, the locking mechanism **54''** may include a top opening **53**. A first set of terminals **66''** may be disposed within the first protrusion **64''**.

[0297] The first protrusion **64''** may be imparted with a first protrusion shape. In some embodiments, the first protrusion shape may be substantially cylindrical with a partially flattened top. The first protrusion shape may be imparted with a substantially circular cross-section where the circular cross-section is partially flattened on top. In other embodiments, the first protrusion **64''** may be imparted with other shapes, for example, a rectangular or square shape, a triangular shape,

a trapezoidal shape, and the like, a combination of shapes, or an irregular shape.

[0298] The first protrusion **64''** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the first protrusion **64''** may include a width, a height, and/or a depth. In some embodiments, the width of the first protrusion **64''** may be about 63 mm. In some embodiments, the height of the first protrusion **64''** may be about 56 mm. In some embodiments, the depth of the first protrusion **64''** may be about 34.5 mm.

[0299] Turning to FIGS. **20A-20D**, the fifth embodiment of the EV lockout device **700** may include a housing **702**. The housing **702** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the housing **702** may be made of ABS polymer plastic. In some embodiments, the housing **702** may be formed using injection molding or a 3D printer.

[0300] The housing **702** may include a base **704** defined by a front side **706** and a back side **708**. The base **704** may also have a top portion **710** and a bottom portion **712**, opposite the top portion **710**. In some embodiments, the top portion **710** terminates at a top edge **711**. The base **704** may have an outer edge **714** that defines a shape of the base **704**. In some embodiments, the shape of the base **704** may be substantially a square, a rectangle, a circle, an oval, a triangle, a trapezoid, and the like. In some embodiments, the shape of the base **704** may be imparted with round edges as shown in FIGS. **20A**, **20B**, and **20D**. Combinations of shapes and irregular shapes are also contemplated.

[0301] In some embodiments, the base **704** may be imparted with a substantially circular shape with a flattened top portion **710** and a rounded bottom portion **712** as shown in FIGS. **20B** and **20D**. In some embodiments, the base may be imparted with a base diameter (or width) **718**. In some embodiments, the base diameter (or width) **718** may be 58 mm, or about 58 mm.

[0302] The base **704** may be imparted with a base height **720**. In some embodiments, the base height **720** may be 50 mm, or about 50 mm. The base **704** also may be imparted with a base thickness (not shown). In some embodiments, the base thickness may be 3 mm, or about 3 mm.

[0303] The front side **706** of the base **704** may include a first flange **724** protruding outwardly therefrom. The first flange **724** may have a top flange portion **725** and a bottom flange portion **727**. The first flange **724** may also have a first flange wall **730** that is imparted with a first flange thickness. In some embodiments, the first flange thickness may be 1.5 mm, or about 1.5 mm.

[0304] The first flange wall **730** may define a first flange shape. The first flange shape may be any size and shape suitable to engage with the first protrusion **64''** when the EV lockout device **700** is placed on or otherwise in communication with the third EV charge plug **50''**. In some embodiments, the first flange shape may be the same or substantially the same as the shape of the first protrusion **64''** of the third EV charge plug **50''**. In some embodiments, the first flange shape may be substantially cylindrical with a flattened top flange portion **725**. In some embodiments, the first flange shape may be imparted with a substantially circular cross-section, where the circular cross-section has a partially flattened top flange portion **725** as shown in FIG. **20B**.

[0305] The first flange **724** may be imparted with a first set of dimensions. In some embodiments, the first set of dimensions of the first flange **724** may be complimentary to the set of dimensions for the first protrusion **64''** so that the first flange **724** can be placed into and fit within the first protrusion **64''**. In some instances, a friction fit between the first flange wall **730** and the inner wall **63** of the first protrusion **64''** may prevent the housing **702** from disengaging from the third EV charge port **50''**.

[0306] In some embodiments, the first set of dimensions of the first flange **724** may include a first flange width **732**, a first flange height **733** (see FIG. **20B**), and a first flange depth **734** (see FIG. **22C**). In some embodiments, the first flange width **732** may be the same as or less than the width of the first protrusion **64''** of the third EV charge plug **50''**. For example, in some embodiments, the first flange width **732**, may be 48 mm, or about 48 mm.

[0307] In some embodiments, the first flange height **733** may be the same as or less than the height of the first protrusion **64''** of the third EV charge plug **50''**. For example, in some embodiments, the

first flange height **733**, may be 40.51 mm, or about 40.51 mm.

[0308] The first flange depth **734** may be the same as or different from the depth of the first protrusion **64"** of the third EV charge plug **50"**. In some embodiments, the first flange depth **734** is the same as, substantially the same as, or less than the depth of the first protrusion **64"** of the third EV charge plug **50"**. For example, in some embodiments, the first flange depth **734** may be 30 mm, or about 30 mm.

[0309] The back side **708** of the base **704** may include a clip support unit **750** protruding outwardly therefrom. The clip support unit **750** may be defined by a bottom support **752**, a first side **754a** and a second side **754b**. In some embodiments, the first side **754a** may be substantially parallel to the second side **754b**. In some embodiments, the second side **754b** may be a mirror image of the first side **754a**.

[0310] The first side **754a** may include a first side plate **756a** having a first bottom end **758a** and a first top end **760a**. The first side plate **756a** may be attached to the bottom support **752** on the first bottom end **758a**. The second side **754b** may include a second side plate **756b** having a second bottom end **758b** and a second top end **760b**. The second side plate **756b** may be attached to the bottom support **752** on the second bottom end **758b**.

[0311] The bottom support **752** may connect the first bottom end **758a** of the first side plate **756a** to the second bottom end **758b** of the second side plate **756b**. A gap **762** is formed between the first side plate **756a** and the second side plate **756b** as shown in FIG. **20D**.

[0312] A first hole **764a** may be disposed in and extend entirely through the first side plate **756a** between the first top end **760a** and the first bottom end **758a**. A second hole **764b** may also be disposed in and extend entirely through the second side plate **756b** between the second top end **760b** and the second bottom end **758b**. The second hole **764b** may be disposed in the second side plate **756b** at the same location as the first hole **764a** is disposed in the first side plate **756a** so that the first hole **764a** and the second hole **764b** are substantially aligned. The first and second holes **764a**, **764b** may be any size and shape suitable for a shackle or shank of a lock (e.g., shackle **14** of the lock **10**) to pass through. In some embodiments, the first and second holes **764a**, **764b** may be imparted with a substantially oval shape. In other embodiments, the first and second holes **764a**, **764b** may be imparted with a substantially circular shape.

[0313] The first side **754a** may further include a first clip support **766a** on the first top end **760a**, the second side **754b** may include a second clip support **766b** on the second top end **760b**. A first vertical support **768a** may extend from the first clip support **766a** upward and past the top edge **711** of the top portion **710** of the base **704**. A second vertical support **768b** may extend from the second clip support **766b** upward and past the top edge **711** of the top portion **710** of the base **704**. In some embodiments, the first vertical support **768a** and the second vertical support **768b** may be mirror images of each other.

[0314] A clip connector **770** may extend between a first top **772a** of the first vertical support **768a** and a first top **772b** of the second vertical support **768b** thereby connecting the first vertical support **768a** to the second vertical support **768b**. The clip connector **770** may also extend from the first vertical support **768a** to the second vertical support **768b** in such a manner that an opening **774** is defined by the first vertical support **768a**, the second vertical support **768b**, the top edge **711** of the base **704**, and the clip connector **770**. In some embodiments, the clip connector **770** may be imparted with a cylindrical or rod shape.

[0315] The clip support unit **750** may be designed to engage with one or more embodiments of a clip (e.g., **200**, **400**, **800**, **1000**) disclosed herein. In some embodiments, the first and second clip supports **766a**, **766b** may be designed to engage or otherwise be in communication with first and second support legs (e.g., **822a**, **822b**), respectively, of a clip (e.g., **800**). The clip connector **770** may be designed to engage with a coupling member (e.g., **816**) of the clip such that the coupling member can rotate about the clip connector **770** in order to raise and/or lower a first end (e.g., **804**) of a first elongated member (e.g., **802**) of the clip. In addition, the gap **762** disposed between the

first and second side plates **756a**, **756b** of the clip support unit **750** may be designed to allow a second elongated member (e.g., **828**) of the clip to move in a substantially upwardly or downwardly direction between the first and second side plates **756a**, **756b**. Furthermore, the first and second holes **764a**, **764b** of the clip support unit **750** may be designed to substantially align with a lock through hole (e.g., **836**) of the clip to enable the shackle of a lock (e.g., shackle **14** of lock **10**) to pass through the housing **702** and clip and secure same to an EV charge port or an EV charge plug (e.g., **50**”).

[0316] The fifth embodiment of the EV lockout device **700** may include a third embodiment of a clip **800** designed for use therewith. The clip **800** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the clip **800** may be made of polypropylene. In some embodiments, the clip **800** may be formed using injection molding or a 3D printer.

[0317] As shown in FIGS. **21A** and **21B**, the clip **800** may be provided in the form of a body having a first elongated member **802** defining a first end **804** and a second end **806**, opposite the first end **804**. The first elongated member **802** may have a first elongated side **807a** and a second elongated side **807b**, whereby the first and second elongated sides **807a**, **807b** may extend between the first end **804** and the second end **806** of the first elongated member **802**. The first elongated member **802** may further be defined by a top side **808** and a bottom side **810**.

[0318] The first elongated member **802** may terminate at a hook portion **812** that extends downwardly from the bottom side **810** at the first end **804**. In some embodiments, the hook portion **812** may be designed to engaged with the locking mechanism **54**” (e.g., the top opening **53**) of the third EV charge plug **50**”. In some embodiments, the hook portion **812** may be imparted with a substantially triangular cross-sectional profile. In other embodiments, the hook portion **812** may be imparted with a different shape.

[0319] The first elongated member **802** may further include a rail **814** that extends along the top side **808** of the first elongated member **802**. The rail **814** may be disposed along a center of and be coextensive with the top side **808** between the first elongated side **807a** and the second elongated side **807b**. In some embodiments, the rail **814** may provide structural support or additional rigidity to the first elongated member **802**.

[0320] The first elongated member **802** may further include a coupling member **816** protruding downwardly from the bottom side **810** thereof. The coupling member **816** may be disposed on the bottom side **810** of the first elongated member **802** adjacent the second end **806**. In some embodiments, the bottom side **810** of the first elongated member **802** together with the coupling member **816** may define a “C” or “U” shape nook **817** as shown in FIG. **21B**. The coupling member **816** may include a projection **818** on an end portion of the coupling member **816**. In some embodiments, the projection **818** may extend upwardly toward the bottom side **810** of the first elongated member **802**. In some embodiments, the nook **817** and the projection **818** may be designed to keep the clip **800** engaged with the housing **702** when placed thereon (see, e.g., FIG. **22C**).

[0321] In some embodiments, the first elongated member **802** may be made of a flexible, but sturdy material, so that the hook portion **812** can slide up and over the top side **65** of the first protrusion **64**” and into the top opening **53** of the third EV charge plug **50**” when the clip **800** is attached to or engaged with the housing **702** of the EV lockout device **700**, but is not able to be easily tampered with or broken. In some embodiments, the first elongated member **802** also may be imparted with a length that is suitable for the hook portion **812** to slide up and over the top side **65** of the first protrusion **64**” and into the top opening **53** of the third EV charge plug **50**” and secure the EV lockout device **700** to the first protrusion **64**” of the third EV charge plug **50**”, when the clip **800** is attached or engaged with the housing **702** of the EV lockout device **700**.

[0322] The second end **806** of the first elongated member **802** may be connected to or attached to a platform portion **820**. The platform portion **820** may include a first support leg **822a** disposed on a

first platform side **824a** of the platform portion **820** and a second support leg **822b** disposed on a second platform side **824b** of the platform portion **820**. In some embodiments, the second support leg **822b** may be substantially parallel to and a mirror image of the first support leg **822a**. The first and second support legs **822a**, **822b** may extend from a front end **826** of the platform portion **820** on an angle downward and away from the front end **826**. In some embodiments, the angle at which the first and second support legs **822a**, **822b** may extend downwardly from the front end **826** of the platform portion **820** may be 45 degrees or about 45 degrees. In other embodiments, the angle may be greater than or less than 45 degrees. In some embodiments, the first and second support legs **822a**, **822b** may be made of a flexible, but sturdy material, so that the first and second support legs **822a**, **822b** can compress and expand without breaking. In some embodiments, the first and second support legs **822a**, **822b** may be springs or a flexible material.

[0323] In some embodiments, the platform portion **820** may be imparted with a substantially rectangular shape. In other embodiments, the platform portion **820** may be imparted with another shape (e.g., elliptical) or an irregular shape. In some embodiments, the platform portion **820** may be imparted with a length and a width and/or a surface area that is large enough for a human finger or thumb to engage with and press downward on the platform portion **820**. For example, in some embodiments, the platform portion **820** may be imparted with a length of 13.88 mm, or about 13.88 mm, and a width of 22.5 mm, or about 22.5 mm.

[0324] The clip **800** may further include a second elongated member **828** that extends downwardly from a center portion **830** of the platform portion **820**. The second elongated member **828** may have a top end **832** and a bottom end **834** and may be connected or attached to the center portion **830** of the platform portion **820** at the top end **832** of the second elongated member **828**. In some embodiments, the second elongated member **828** may extend downwardly from the platform portion **820** such that the first elongated member **802** and second elongated member **828** are disposed substantially perpendicular to one another as shown in FIG. 21B. In some embodiments, the second elongated member **828** may be imparted with a length such that a space **835** exists between the bottom support **752** of the housing **702** of the EV lockout device **700** and the bottom end **834** of the second elongated member **828** as shown in FIGS. 22B and 22C.

[0325] A lock through hole **836** may be disposed in and extend entirely through the second elongated member **828** between the top end **832** and the bottom end **834**. In some embodiments, the lock through hole **836** may be disposed adjacent the bottom end **834**. The lock through hole **836** may be imparted with a size and shape that is suitable for a shackle or shank of a lock (e.g., shackle **14** of lock **10**) to pass through. In some embodiments, the lock through hole **836** may be the same shape and size as the first and second holes **764a**, **764b** of the clip support unit **750** of the housing **702**. In some embodiments, the lock through hole **836** may be the same shape, but a different size as the first and second holes **764a**, **764b** of the clip support unit **750** of the housing **702**. In other embodiments, the lock through hole **836** may be a different size and/or shape as the first hole **764a** or the second hole **764b** of the clip support unit **750** of the housing **702**. In some embodiments, the lock through hole **836** may be disposed at a location in the second elongated member **828** such that the lock through hole **836** substantially aligns with the first hole **764a** and the second hole **764b** of the clip support unit **750** when the clip **800** is attached to or engages with the housing **702** in a first clip position (see, e.g., FIG. 22C).

[0326] As shown in FIGS. 22A-22C, the clip **800** may attach to or otherwise engage with the housing **702**. When in the first clip position, the clip **800** may be placed on the housing **702** such that the coupling member **816**, the bottom side **810** of the first elongated member **802**, and/or the projection **818** of the clip **800** may engage with the clip connector **770** of the housing **702** as best seen in FIG. 22C. The clip connector **770** may rest within the nook **817** when in the first clip position, and the coupling member **816** may extend through the opening **774** of the housing **702** as shown in FIG. 22C. In addition, when in the first clip position, the first support leg **822a** may engage with the first clip support **766a** of the clip support unit **750**, and the second support leg

822b may engage with the second clip support **766b** of the clip support unit **750**. The first elongated member **802** of the clip **800** may extend over the top flange portion **725** of the first flange **724**. The second elongated member **828** may be disposed within the gap **762** between the first side **754a** and the second side **754b** of the clip support unit **750** as shown in FIG. 22B. Further, when in the first clip position, the lock through hole **836** of the clip **800** may substantially align with the first hole **764a** of the first side **754a** and the second hole **764b** of the second side **754b** of the clip support unit **750**.

[0327] As discussed above, when in the first clip position, the first and second support legs **822a**, **822b** may engage with the first and second clip supports **766a**, **766b**, respectively. In addition, when in the first clip position, the lock through hole **836** of the second elongated member **828** of the clip **800** may substantially align with the first hole **764a** of the first side **754a** and the second hole **764b** of the second side **754b** of the clip support unit **750** of the housing **702**.

[0328] When the EV lockout device **700** is being placed on or is being removed from the third EV charge plug **50''**, the clip **800** may move to a second clip position (not shown). When moving into the second clip position, the first end **804** of the first elongated member **802** may move in a substantially upward direction. In some embodiments, the clip **800** may move to the second clip position in response to the platform portion **820** being pressed downward so that the first and second support legs **822a**, **822b** compress (e.g., the angle between the platform portion **820** and one or more of the first and second support legs **822a**, **822b** decreases). In other embodiments, the clip **800** may move to the second clip position in response to the hook portion **812** of the first elongated member **802** moving up and over the top side **65** of the first protrusion **64''** of the third EV charge plug **50''**. In the second clip position, the first end **804** of the first elongated member **802** may raise up in order for the hook portion **812** to slide over the top side **65** of the first protrusion **64''** and into the top opening **53** of the third EV charge plug **50''**. As the first end **804** of the first elongated member **802** raises upward, the coupling member **816** and the projection **818** may move or rotate about the clip connector **770** of the housing **702** in a first direction. In some embodiments, as the first end **804** of the first elongated member **802** raises upward, the second elongated member **828** may move in a substantially downward direction such that the space **835** between the bottom end of the second elongated member **828** and the bottom support **752** of the clip support unit **750** decreases. In some embodiments, as the second elongated member **828** moves in the substantially downward direction, the lock through hole **836** of the second elongated member **828** may become out of alignment with the first hole **764a** and the second hole **764b** of the clip support unit **750**.

[0329] If the EV lockout device **700** is being placed into the third EV charge plug **50''**, the clip **800** may move from the first clip position to the second clip position. As the clip **800** is moving from the first clip position to the second clip position, the housing **702** may engage with the third EV charge plug **50''**. In some embodiments, the first flange **724** of the housing **702** may engage with the inner wall **63** of the first protrusion **64''** of the third EV charge plug **50''** such that the first protrusion **64''** surrounds all or a portion of the first flange **724**.

[0330] After moving into the second clip position in response to the platform portion **820** being depressed, the clip **800** may move back into the first clip position upon release of the platform portion **820**. As the clip **800** moves back into the first clip position, the first end **804** of the first elongated member **802** may lower and the coupling member **816** and the projection **818** may move or rotate about the clip connector **770** of the housing **702** in a direction opposite the first direction. In addition, as the clip **800** moves back in the first clip position, the second elongated member **828** may move in a substantially upward direction as the first and second support legs **822a**, **822b** decompress (e.g., the angle between the platform portion **820** and one or more of the first and second support legs **822a**, **822b** increases). If the EV lockout device **700** has been placed onto the third EV charge plug **50''**, the hook portion **812** may be disposed in the top opening **53** on the top side **65** of the first protrusion **64''**, when the clip **800** returns to the first clip position. Additionally, when the clip **800** moves back into the first clip position, the first hole **764a** of the

clip support unit **750**, the lock through hole **836** of the second elongated member **828**, and the second hole **764b** of the clip support unit **750** may be substantially aligned such that the shackle **14** of lock **10** may be threaded through the first hole **764a**, the lock through hole **836**, and the second hole **764b**, or vice versa. The shackle **14** may then be securely attached to the body **12** of the lock **10**, thereby placing the EV lockout device **700** in a locked position as shown in FIG. 23.

[0331] When in the locked position, the clip **800** cannot fully move to the second clip position (e.g., the hook portion **812** cannot be raised high enough to slide out of the top opening **53** and back over the top side **65** of the first protrusion **64"** of the third EV charge plug **50"**) as the shackle **14** prevents the second elongated member **828** of the clip **800** from moving sufficiently downward. To remove the EV lockout device **700** from the third EV charge plug **50"**, the shackle **14** of the lock **10** may be disengaged from the body **12** of the lock **10** and removed from the clip support unit **750** of the housing **702** by threading the shackle **14** through the second hole **764b**, the lock through hole **836**, and the first hole **764a**, or vice versa. Once the lock **10** is removed from the EV lockout device **700**, the platform portion **820** of the clip **800** may be depressed causing the clip **800** to move from the first clip position to the second clip position in which the first and second support legs **822a**, **822b** compress and the hook portion **812** is raised. The hook portion **812** may then be moved up and away from the top opening **53** as the EV lockout device **700** is slid off or otherwise removed from the third EV charge plug **50"** (e.g., the first flange **724** is removed from the first protrusion **64"**).

[0332] FIG. 24 depicts a partially exploded view of a lock **10**, a sixth embodiment of an EV lockout device **900**, and a third electric vehicle (EV) charge port **20"**.

[0333] The third EV charge port **20"** may be disposed inside of, on, or otherwise associated with an electric vehicle (not shown). In some embodiments, the third EV charge port **20"** may be provided in the form of a North American Charging Standard (NACS) or Tesla socket.

[0334] The third EV charge port **20"** may include a port housing **22"** having a locking mechanism aperture **24'**. The third EV charge port **20"** may include a locking mechanism **30"** disposed within an inner wall **31** of the locking mechanism aperture **24'**. The locking mechanism **30"** may include a bottom opening **35** disposed on a bottom portion of the inner wall **31** of the locking mechanism aperture **24'**.

[0335] A first set of terminals receptors **36"** may be disposed within the locking mechanism aperture **24'**.

[0336] The locking mechanism aperture **24'** may be imparted with a first shape. In some embodiments, the first shape may be partially cylindrical. In other embodiments, the locking mechanism aperture **24'** may be imparted with other shapes, for example, a rectangular or square shape, a triangular shape, a trapezoidal shape, and the like, a combination of shapes, or an irregular shape.

[0337] The locking mechanism aperture **24** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the locking mechanism aperture **24'** may include a center width, a center height, and/or a depth. In some embodiments, the center width of the locking mechanism aperture **24'** may be about 40.35 mm. In some embodiments, the center height of the locking mechanism aperture **24'** may be about 36.14 mm. In some embodiments, the depth of the locking mechanism aperture **24'** may be about 49.86 mm.

[0338] Turning to FIGS. 25A-25D, the sixth embodiment of the EV lockout device **900** may include a housing **902**. The housing **902** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the housing **902** may be made of ABS polymer plastic. In some embodiments, the housing **902** may be formed using injection molding or a 3D printer.

[0339] The housing **902** may include a base **904** defined by a front side **906** and a back side **908**. The base **904** may also have a top portion **910** and a bottom portion **912**, opposite the top portion **910**. In some embodiments, the bottom portion **912** may have a bottom edge **913**.

[0340] The base **904** may have an outer edge **914** that defines a shape of the base **904**. In some embodiments, the shape of the base **904** may be substantially a square, a rectangle, a circle, an oval, a triangle, a trapezoid, and the like. In some embodiments, the shape of the base **904** may be imparted with round edges as shown in FIGS. 25A, 25C, and 25D. Combinations of shapes and irregular shapes are also contemplated.

[0341] In some embodiments, the base **904** may be imparted with a shape that is substantially oblong or a substantially boat shape as shown in FIGS. 25C and 25D. In some embodiments, the base **904** may be imparted with a base width **916**. In some embodiments, the base width **916** may be 40.38 mm, or about 40.38 mm.

[0342] The base **904** may be imparted with a base height **920** (see FIG. 25B). In some embodiments, the base height **920** may be 71.14 mm, or about 71.14 mm. The base **904** also may be imparted with a base thickness (not shown). In some embodiments, the base thickness may be 3 mm, or about 3 mm.

[0343] The base **904** may have a clip opening **921**. In some embodiments, the clip opening **921** may be disposed in and extend entirely through the base **904** within a first flange wall **930** of a first flange **924** (discussed in more detail below) as shown in FIG. 25C. The clip opening **921** may be imparted with any size or shape suitable for a first end **1004** of a first elongated member **1002** of a fourth embodiment of a clip **1000**, discussed below, and/or a fin portion **1037** of a second elongated member **1028** of the clip **1000** to fit partially or fully therethrough. In some embodiments, the clip opening **921** may have a substantially rectangular shape as shown in FIGS. 25C and 25D.

[0344] The bottom portion **912** of the base **904** may have a divot opening **922** disposed in and extend entirely through a center area of the bottom portion **912**. In some embodiments, the divot opening **922** may be disposed below the clip opening **921**. The divot opening **922** may be defined by a first divot side **923a**, a second divot side **923b**, a bottom divot side **923c**, and a top divot side **923d**. In some embodiments, the divot opening **1122** may be imparted with a substantially rectangular shape as shown in FIGS. 25C and 25D.

[0345] The front side **906** of the base **904** may include the first flange **924**. In some embodiments, the first flange **924** may be elongated as shown in FIG. 25B. The first flange **924** may have a top flange portion **925** and a bottom flange portion **927**. The first flange **924** may also have an inner side **931**.

[0346] In some embodiments, a first flange aperture **929** may be disposed in the bottom flange portion **927** as shown in FIGS. 25B, 27B, and 27D. The first flange aperture **929** may be imparted with any size or shape suitable for a first hook portion **1012a** and/or a second hook portion **1012b** of the clip **1000**, discussed in more detail below, to fit partially or fully therethrough (see, e.g., FIG. 27D).

[0347] The first flange **924** may have a first flange wall **930** that is imparted with a first flange thickness. In some embodiments, the first flange thickness may be 1.6 mm, or about 1.6 mm.

[0348] The first flange wall **930** may define a first flange shape. The first flange shape may be any size and shape suitable to engage with the locking mechanism aperture **24'** when the EV lockout device **900** is placed on or otherwise in communication with the third EV charge port **20''**. In some embodiments, the first flange shape may be the same or substantially the same as the shape of the locking mechanism aperture **24'** of the third EV charge port **20''**. In some embodiments, the first flange shape may be partially cylindrical.

[0349] The first flange **924** may be imparted with a first set of dimensions. In some embodiments, the first set of dimensions of the first flange **924** may be complimentary to the set of dimensions for the locking mechanism aperture **24'** of the third EV charge port **20''** so that the first flange **924** can be placed into and fit within the locking mechanism aperture **24'** as shown in FIG. 28. In some instances, a friction fit between the first flange wall **930** and the inner wall **31** of the locking mechanism aperture **24'** may prevent the housing **902** from disengaging from the third EV charge port **20''**.

[0350] In some embodiments, the first set of dimensions of the first flange **924** may include a first flange center width **932**, a first flange center height **933** (see FIG. 25C), and a first flange depth **934** (see FIG. 27D). In some embodiments, the first flange center width **932** may be the same as or less than the center width of the locking mechanism aperture **24'** of the third EV charge port **20''**. For example, in some embodiments, the first flange center width **932**, may be 35.56 mm, or about 35.56 mm, and the first flange center height **933** may be 31.24 mm, or about 31.24 mm.

[0351] The first flange depth **934** may be the same as or different from the depth of the locking mechanism aperture **24'** of the third EV charge port **20''**. In some embodiments, the first flange depth **934** is the same as, substantially the same as, or greater than the depth of the locking mechanism aperture **24'** of the third EV charge port **20''**. For example, in some embodiments, the first flange depth **934** may be 69 mm, or about 69 mm.

[0352] The back side **908** of the base **904** may include a clip support unit **950** and a bottom clip support **951** protruding outwardly therefrom. The clip support unit **950** may be defined by a top support **953**, a first side **954a**, and a second side **954b**. In some embodiments, the first side **954a** may be substantially parallel to the second side **954b**. In some embodiments, the second side **954b** may be a mirror image of the first side **954a**.

[0353] The first side **954a** may include a first side plate **956a** having a first bottom end **958a** and a first top end **960a**. The first side plate **956a** may be connected or attached to the top support **953** on the first top end **960a**. The second side **954b** may include a second side plate **956b** having a second bottom end **958b** and a second top end **960b**. The second side plate **956b** may be connected or attached to the top support **953** on the second top end **960b**.

[0354] The top support **953** may connect the first side plate **956a** to the second side plate **956b**. A gap **962** is formed between the first side plate **956a** and the second side plate **956b** as shown in FIG. 25D.

[0355] A first hole **964a** may be disposed in and extend entirely through the first side plate **956a** between the first top end **960a** and the first bottom end **958a**. A second hole **964b** may also be disposed in and extend entirely through the second side plate **956b** between the second top end **960b** and the second bottom end **958b**. The second hole **964b** may be disposed in the second side plate **956b** at the same location as the first hole **964a** is disposed in the first side plate **956a** so that the first hole **964a** and the second hole **964b** are substantially aligned. The first and second holes **964a**, **964b** may be any size and shape suitable for a shackle or shank of a lock (e.g., shackle **14** of the lock **10**) to pass through. In some embodiments, the first and second holes **964a**, **964b** may be imparted with a substantially oval shape. In other embodiments, the first and second holes **964a**, **964b** may be imparted with a substantially circular shape.

[0356] In some embodiments, the bottom clip support **951** may be imparted with a substantially rectangular shape as shown in FIG. 27B. The bottom clip support **951** may include a first vertical support **968a** and a second vertical support **968b**. The first vertical support **968a** may extend from the bottom clip support **951** upward toward the top portion **910** of the housing **902** and the second vertical support **968b** may extend from the bottom clip support **951** upward toward the top portion **910** of the housing **902**. In some embodiments, the first vertical support **968a** and the second vertical support **968b** may be mirror images of each other. The first and second vertical supports **968a**, **968b** may also be disposed on the back side **908** of the housing **902** adjacent the divot opening **922** such that the first vertical support **968a** is disposed adjacent the first divot side **923a** and the second vertical support **968b** is disposed adjacent the second divot side **923b** of the divot opening **922**.

[0357] A clip connector **970** may extend between the first vertical support **968a** and the second vertical support **968b** thereby connecting the first vertical support **968a** to the second vertical support **968b**. The clip connector **970** may also extend across the divot opening **922**. In some embodiments, the clip connector **970** may be imparted with a cylindrical or rod shape.

[0358] The clip support unit **950** and the bottom clip support **951** may be designed to engage with

one or more embodiments of a clip (e.g., **200**, **400**, **800**, **1000**) disclosed herein. In some embodiments, the bottom clip support **951** may be designed to engage or otherwise be in communication with first and second support legs (e.g., **1022a**, **1022b**) of a clip (e.g., **1000**). The clip connector **970** may be designed to engage with a coupling member (e.g., **1016**) of the clip such that the coupling member can rotate about the clip connector **970** in order to raise and/or lower a first end (e.g., **1004**) of a first elongated member (e.g., **1002**) of the clip. In addition, the gap **962** disposed between the first and second side plates **956a**, **956b** of the clip support unit **950** may be designed to engage with a second elongated member (e.g., **1028**) of the clip and allow the second elongated member of the clip to move in a substantially upwardly or downwardly direction between the first and second side plates **956a**, **956b**. Furthermore, the first and second holes **964a**, **964b** of the clip support unit **950** may be designed to substantially align with a lock through hole (e.g., **1036**) of the clip to enable the shackle of a lock (e.g., shackle **14** of lock **10**) to pass through the housing **902** and clip and secure same to an EV charge port (e.g., **20"**) or an EV charge plug. [0359] The sixth embodiment of the EV lockout device **900** may include a fourth embodiment of a clip **1000** designed for use therewith. The clip **1000** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the clip **1000** may be made of polypropylene. In some embodiments, the clip **1000** may be formed using injection molding or a 3D printer.

[0360] As shown in FIGS. **26A** and **26B**, the clip **1000** may be provided in the form of a body having a first elongated member **1002** defining a first end **1004** and a second end **1006**, opposite the first end **1004**. The first elongated member **1002** may have a first elongated side **1007a** and a second elongated side **1007b**, whereby the first and second elongated sides **1007a**, **1007b** may extend between the first end **1004** and the second end **1006** of the first elongated member **1002**. The first elongated member **1002** may further be defined by a top side **1008** and a bottom side **1010**.

[0361] The first elongated member **1002** may terminate at a first hook portion **1012a** and a second hook portion **1012b** that extend downwardly from the bottom side **1010** of the first elongated side **1007a** at the first end **1004**. In some embodiments, the first and second hook portions **1012a**, **1012b** may be designed to engage with the locking mechanism **30"** (e.g., bottom opening **35**) of the third EV charge port **20"** of FIG. **24**. In some embodiments, the first and second hook portions **1012a**, **1012b** may be imparted with a substantially triangular cross-sectional profile. In other embodiments, the first and second hook portions **1012a**, **1012b** may be imparted with a different shape.

[0362] The first elongated member **1002** may further include a coupling member **1016** protruding downwardly from the bottom side **1010** thereof. The coupling member **1016** may be disposed on the bottom side **1010** of the first elongated member **1002** adjacent the second end **1006**. In some embodiments, the bottom side **1010** of the first elongated member **1002** together with the coupling member **1016** may define a "C" or "U" shape nook **1017** as shown in FIG. **26B**. The coupling member **1016** may include a projection **1018** on an end portion of the coupling member **1016**. In some embodiments, the projection **1018** may extend upwardly toward the bottom side **1010** of the first elongated member **1002**. In some embodiments, the nook **1017** and the projection **1018** may be designed to keep the clip **1000** engaged with the housing **902** when placed thereon (see, e.g., FIG. **27D**).

[0363] In some embodiments, the first elongated member **1002** may be made of a flexible, but sturdy material so that the first and second hook portions **1012a**, **1012b** can slide up and over the inner wall **31** of the locking mechanism aperture **24** and into the bottom opening **35** of the locking mechanism **30"** of the third EV charge port **20"** when the clip **1000** is attached to or engaged with the housing **902** of the EV lockout device **900**), but is not able to be easily tampered with or broken. In some embodiments, the first elongated member **1002** may be imparted with a length that is suitable for first and second hook portions **1012a**, **1012b** to slide up and over the

inner wall **31** of the locking mechanism aperture **24'** of the third EV charge port **20''** and secure the EV lockout device **900** to the locking mechanism **30''**, when the clip **1000** is attached or engaged with the housing **902** of the EV lockout device **900**.

[0364] The second end **1006** of the first elongated member **1002** may be connected to or attached to a platform portion **1020**. The platform portion **1020** may include a first support leg **1022a** disposed on a first platform side **1024a** of the platform portion **1020** and a second support leg **1022b** disposed on a second platform side **1024b** of the platform portion **1020**. In some embodiments, the second support leg **1022b** may be substantially parallel to and a mirror image of the first support leg **1022a**. The first and second support legs **1022a**, **1022b** may extend from a front end **1026** of the platform portion **1020** on an angle downward and away from the front end **1026**. In some embodiments, the angle at which the first and second support legs **1022a**, **1022b** may extend downwardly from the front end **1026** of the platform portion **1020** may be 45 degrees or about 45 degrees. In other embodiments, the angle may be greater than or less than 45 degrees. In some embodiments, the first and second support legs **1022a**, **1022b** may be made of a flexible, but sturdy material, so that the first and second support legs **1022a**, **1022b** can compress and expand without breaking. In some embodiments, the first and second support legs **1022a**, **1022b** may be springs or a flexible material.

[0365] In some embodiments, the platform portion **1020** may be imparted with a substantially rectangular shape. In other embodiments, the platform portion **1020** may be imparted with another shape (e.g., elliptical) or an irregular shape. In some embodiments, the platform portion **1020** may be imparted with a length and a width and/or a surface area that is large enough for a human finger or thumb to engage with and press downward on the platform portion **1020**. For example, in some embodiments, the platform portion **1020** may be imparted with a length of 13.88 mm, or about 13.88 mm, and a width of 22.5 mm, or about 22.5 mm.

[0366] The clip **1000** may further include a second elongated member **1028** that extends upwardly from the top side **1008** of the second end **1006** of the first elongated member **1002**. The second elongated member **1028** may extend upwardly from a center section **1030** of the first elongated member **1002** between the first elongated side **1007a** and the second elongated side **1007b**. The second elongated member **1028** may have a top end **1032** and a bottom end **1034** and may be connected or attached to the center section **1030** of the first elongated member **1002** at the bottom end **1034** of the second elongated member **1028**. In some embodiments, the second elongated member **1028** may extend upwardly from the first elongated member **1002** such that the first elongated member **1002** and second elongated member **1028** are disposed substantially perpendicular to one another as shown in FIG. 26B. In some embodiments, the second elongated member **1028** may be imparted with a length such that a space **1035** exists between the top support **953** of the housing **902** of the EV lockout device **900** and the top end **1032** of the second elongated member **1028** when the clip **1000** is attached to or engaged with the housing **902** as shown in FIGS. 27C and 27D.

[0367] In some embodiments, the second elongated member **1028** may be imparted with a fin portion **1037** that is coextensive with and extends from the second elongated member **1028** toward the first end **1004** of the first elongated member **1002** (see, e.g., FIG. 26B). In some embodiments, the fin portion **1037** may be imparted with a substantially triangular shape. In some embodiments, a bottom **1037** of the fin portion **1037** may be connected or attached to the first elongated member **1002** along the center section **1030** of the first elongated member.

[0368] A lock through hole **1036** may be disposed in and extend entirely through the second elongated member **1028** between the top end **1032** and the bottom end **1034**. In some embodiments, the lock through hole **1036** may be disposed adjacent the top end **1032**. The lock through hole **1036** may be imparted with a size and shape that is suitable for a shackle or shank of a lock (e.g., shackle **14** of lock **10**) to pass through. In some embodiments, the lock through hole **1036** may be the same shape and size as the first and second holes **964a**, **964b** of the clip support

unit **950** of the housing **902**. In some embodiments, the lock through hole **1036** may be the same shape, but a different size as the first and second holes **964a**, **964b** of the clip support unit **950** of the housing **902**. In other embodiments, the lock through hole **936** may be a different size and/or shape as the first hole **964a** or the second hole **964b** of the clip support unit **950** of the housing **902**. In some embodiments, the lock through hole **1036** may be disposed at a location in the second elongated member **1028** such that the lock through hole **1036** substantially aligns with the first hole **964a** and the second hole **964b** of the clip support unit **950** when the clip **1000** is attached to or engages with the housing **902** in a first clip position (see, e.g., FIGS. **27A** and **27D**).

[0369] As shown in FIGS. **27A-27D**, the clip **1000** may attach to or otherwise engage with the housing **902**. When in the first clip position, the clip **1000** may be placed on the housing **902** such that the coupling member **1016**, the bottom side **1010** of the first elongated member **1002**, and/or the projection **1018** of the clip **1000** may engage with the clip connector **970** of the housing **902** as best seen in FIG. **27D**. The clip connector **970** may rest within the nook **1017** when in the first clip position, and the coupling member **1016** may extend through the divot opening **922** of the housing **902** as shown in FIG. **27D**. In addition, when in the first clip position, the first support leg **1022a** and/or the second support leg **1022b** may engage with the bottom clip support **951**. The first elongated member **1002** of the clip **1000** may extend through the clip opening **921** and over the inner side **931** of the first flange **924** of the housing **902**. The first and second hook portions **1012a**, **1012b** may be disposed in the first flange aperture **929** as shown in FIGS. **27A**, **27B**, and **27D**. The second elongated member **1028** may be disposed within the gap **962** between the first side **954a** and the second side **954b** of the clip support unit **950** as shown in FIG. **27C**. Further, when in the first clip position, the lock through hole **1036** of the clip **1000** may substantially align with the first hole **964a** of the first side **954a** and the second hole **964b** of the second side **954b** of the clip support unit **950**.

[0370] As discussed above, when in the first clip position, the first and second support legs **1022a**, **1022b** may engage with the bottom clip support **951**. In addition, when in the first clip position, the lock through hole **1036** of the second elongated member **1028** of the clip **1000** may substantially align with the first hole **964a** of the first side **954a** and the second hole **964b** of the second side **954b** of the clip support unit **950** of the housing **902**.

[0371] When the EV lockout device **100** is being placed on or is being removed from the third EV charge port **20"**, the clip **1000** may move to a second clip position (not shown). When moving into the second clip position, the first end **1004** of the first elongated member **1002** may move in a substantially upward direction. In some embodiments, the clip **1000** may move to the second clip position in response to the platform portion **1020** being pressed downward so that the first and second support legs **1022a**, **1022b** compress (e.g., the angle between the platform portion **1020** and one or more of the first and second support legs **1022a**, **1022b** decreases). In other embodiments, the clip **1000** may move to the second clip position in response to the first and second hook portions **1012a**, **1012b** of the first elongated member **1002** engaging with the inner side **931** of the first flange **924**. In the second clip position, the first end **1004** of the first elongated member **1002** may raise up in order for the first and second hook portions **1012a**, **1012b** to slide over the inner side **931** of the first flange **924** and into the bottom opening **35** of the locking mechanism **30"** of third EV charge port **20"**. As the first end **1004** of the first elongated member **1002** raises upward, the coupling member **1016** and the projection **1018** may move or rotate about the clip connector **970** of the housing **902** in a first direction. In some embodiments, as the first end **1004** of the first elongated member **1002** raises upward, the second elongated member **1028** may move in a substantially downward direction such that the space **1035** between the top end **1032** of the second elongated member **1028** and the top support **953** of the clip support unit **950** increases. In some embodiments, as the second elongated member **1028** moves in the substantially downward direction, the lock through hole **1036** of the second elongated member **1028** may become out of alignment with the first hole **964a** and the second hole **964b** of the clip support unit **950**.

[0372] If the EV lockout device **900** is being placed onto the third EV charge port **20''**, the clip **1000** may move from the first clip position to the second clip position. As the clip **1000** is moving from the first clip position to the second clip position, the housing **902** may engage with the third EV charge port **20''**. In some embodiments, the first flange **924** of the housing **902** may engage with the inner wall **31** of the locking mechanism aperture **24'** of the third EV charge port **20''** such that the locking mechanism aperture **24** surrounds all or a portion of the first flange **924**.

[0373] After moving into the second clip position in response to the platform portion **1020** being depressed, the clip **1000** may move back into the first clip position upon release of the platform portion **1020**. As the clip **1000** moves back into the first clip position, the first end **1004** of the first elongated member **1002** may lower and the coupling member **1016** and the projection **1018** may move or rotate about the clip connector **970** of the housing **902** in a direction opposite the first direction. In addition, as the clip **1000** moves back in the first clip position, the second elongated member **1028** may move in a substantially upward direction as the first and second support legs **1022a**, **1022b** decompress (e.g., the angle between the platform portion **1020** and one or more of the first and second support legs **1022a**, **1022b** increases). If the EV lockout device **900** has been placed on the third EV charge port **20''**, the first and second hook portions **1012a**, **1012b** may be disposed within the first flange aperture **929** and the bottom opening **35** of the locking mechanism **30''**, when the clip **200** returns to the first clip position. Additionally, when the clip **1000** moves back into the first clip position, the first hole **964a** of the clip support unit **950**, the lock through hole **1036** of the second elongated member **1028**, and the second hole **964b** of the clip support unit **950** may be substantially aligned such that the shackle **14** of lock **10** may be threaded through the first hole **964a**, the lock through hole **1036**, and the second hole **964b**, or vice versa. The shackle **14** may then be securely attached to the body **12** of the lock **10**, thereby placing the EV lockout device **900** in a locked position as shown in FIG. **28**.

[0374] When in the locked position, the clip **1000** cannot fully move to the second clip position (e.g., the first and second hook portions **1012a**, **1012b** cannot be raised high enough to move up and out of the bottom opening **35** of the third EV charge port **20''**) as the shackle **14** prevents the second elongated member **1028** of the clip **1000** from moving sufficiently downward. To remove the EV lockout device **900** from the third EV charge port **20''**, the shackle **14** of the lock **10** may be disengaged from the body **12** of the lock **10** and removed from the clip support unit **950** of the housing **902** by threading the shackle **14** through the second hole **964b**, the lock through hole **1036**, and the first hole **964a**, or vice versa. Once the lock **10** is removed from the EV lockout device **900**, the platform portion **1020** of the clip **1000** may be depressed causing the clip **1000** to move from the first clip position to the second clip position in which the first and second support legs **1022a**, **1022b** compress and the first and second hook portions **1012a**, **1012b** are raised. The first and second hook portions **1012a**, **1012b** may then be moved over and away from the bottom opening **35** of the locking mechanism aperture **24'** of the third EV charge port **20''** as the EV lockout device **900** is slid out of or otherwise removed from the third EV charge port **20''** (e.g., the first flange **924** is removed from the locking mechanism aperture **24''**).

[0375] FIG. **29** depicts a partially exploded view of the lock **10**, a seventh embodiment of an EV lockout device **1100**, and a fourth electric vehicle (EV) charge plug **50''**.

[0376] The fourth EV charge plug **50''** may be attached to an EV charging station (not shown). In some embodiments, the fourth EV charge plug **50''** may be provided in the form of a North American Charging Standard (NACS) or Tesla charge plug.

[0377] The fourth EV charge plug **50''** may include a plug housing **52''** having a front side **55''**. A first protrusion **64''** may extend from the front side **55''** of the plug housing **52''**.

[0378] The fourth EV charge plug **50''** may include a locking mechanism **54''** disposed on a top side **65'** of the first protrusion **64''**. In some embodiments, the locking mechanism **54''** may include a top opening **53'**. A first set of terminals **66''** may be disposed with the first protrusion **64''**.

[0379] The first protrusion **64''** may be imparted with a first protrusion shape. In some

embodiments, the first protrusion shape may be partially cylindrical. In other embodiments, the first protrusion **64"** may be imparted with other shapes, for example, a rectangular or square shape, a triangular shape, a trapezoidal shape, and the like, a combination of shapes, or an irregular shape.

[0380] The first protrusion **64"** may be imparted with a set of dimensions (not shown). In some embodiments, the set of dimensions of the first protrusion **64"** may include a center width, a center height, and/or a depth. In some embodiments, the center width of the first protrusion **64"** may be about 39.84 mm. In some embodiments, the center height of the first protrusion **64"** may be about 35.11 mm. In some embodiments, the depth of the first protrusion **64"** may be about 48.36 mm.

[0381] Turning to FIGS. **30A-30D**, the seventh embodiment of the EV lockout device **1100** may include a housing **1102**. The housing **1102** may be made of a polymeric material such as a plastic, metal, or other suitable materials, or combinations thereof. In one embodiment, the housing **1102** may be made of ABS polymer plastic. In some embodiments, the housing **1102** may be formed using injection molding or a 3D printer.

[0382] The housing **1102** may include a base **1104** defined by a front side **1106** and a back side **1108**. The base **1104** may also have a top portion **1110** and a bottom portion **1112**, opposite the top portion **1110**. The base **1104** may have an outer edge **1114** that defines a shape of the base **1104**. In some embodiments, the shape of the base **1104** may be substantially a square, a rectangle, a circle, an oval, a triangle, a trapezoid, and the like. In some embodiments, the shape of the base **1104** may be imparted with round edges as shown in FIGS. **30B** and **30D**. Combinations of shapes and irregular shapes are also contemplated.

[0383] In some embodiments, the base **1104** may be imparted with a substantially circular shape as shown in FIGS. **30B** and **30D**. In some embodiments, the base **1104** may be imparted with a base diameter (or width) **1118**. In some embodiments, the base diameter (or width) **1118** may be 48 mm, or about 48 mm.

[0384] The base **1104** may be imparted with a center base height **1120**. In some embodiments, the center base height **1120** may be 40.92 mm, or about 40.92 mm. The base **1104** also may be imparted with a base thickness (not shown). In some embodiments, the base thickness may be 2.5 mm, or about 2.5 mm.

[0385] The top portion **1110** of the base **1104** may have a divot opening **1122** disposed in a center area of the top portion **1110**. The divot opening **1122** may be defined by a first divot side **1123a**, a second divot side **1123b**, a bottom divot side **1123c**, and a top divot side **1123d**. In some embodiments, the divot opening **1122** may be imparted with a partially rectangular shape with a rounded top as shown in FIGS. **30B** and **30D**.

[0386] The front side **1106** of the base **1104** may include a first flange **1124** protruding outwardly therefrom. The first flange **1124** may have a top flange portion **1125** and a bottom flange portion **1127**. The first flange **1124** may also have a first flange wall **1130** that is imparted with a first flange thickness. In some embodiments, the first flange thickness may be 2 mm, or about 2 mm.

[0387] The first flange wall **1130** may define a first flange shape. The first flange shape may be imparted with any size and shape suitable to engage with the first protrusion **64"** when the EV lockout device **1100** is placed on or otherwise in communication with the fourth EV charge plug **50"**. In some embodiments, the first flange shape may be the same or substantially the same as the shape of the first protrusion **64"** of the fourth EV charge plug **50"**. In some embodiments, the first flange shape may be partially cylindrical.

[0388] The first flange **1124** may be imparted with a first set of dimensions. In some embodiments, the first set of dimensions of the first flange **1124** may be complimentary to the set of dimensions for the first protrusion **64"** so that the first flange **1124** can be placed onto and fit over the first protrusion **64"**. In some instances, a friction fit between the first flange wall **1130** and the first protrusion **64"** may prevent the housing **1102** from disengaging from the charge plug **50"**.

[0389] In some embodiments, the first set of dimensions of the first flange **1124** may include a first flange width **1132**, a first flange center height **1133** (see FIG. **30B**), and a first flange depth **1134**

(sec FIG. 31C). In some embodiments, the first flange diameter **1132** may be the same as or greater than the diameter of the first protrusion **64**" of the fourth EV charge plug **50**". For example, in some embodiments, the first flange width **1132**, may be 41.12 mm, or about 41.12 mm, and the first flange center height **1133** may be 36.16 mm, or about 36.16 mm.

[0390] The first flange depth **1134** may be the same as or different from the depth of the first protrusion **64**" of the fourth EV charge plug **50**". In some embodiments, the first flange depth **1134** is the same as, substantially the same as, or less than the depth of the first protrusion **64**" of the fourth EV charge plug **50**". For example, in some embodiments, the first flange depth **1134** may be 28 mm, or about 28 mm.

[0391] The back side **1108** of the base **1104** may include a clip support unit **1150** protruding outwardly therefrom. The clip support unit **1150** may be defined by a bottom support **1152**, a first side **1154a** and a second side **1154b**. In some embodiments, the first side **1154a** may be substantially parallel to the second side **1154b**. In some embodiments, the second side **1154b** may be a mirror image of the first side **1154a**.

[0392] The first side **1154a** may include a first side plate **1156a** having a first bottom end **1158a** and a first top end **1160a**. The first side plate **1156a** may be attached to the bottom support **1152** on the first bottom end **1158a**. The second side **1154b** may include a second side plate **1156b** having a second bottom end **1158b** and a second top end **1160b**. The second side plate **1156b** may be attached to the bottom support **1152** on the second bottom end **1158b**.

[0393] The bottom support **1152** may connect the first bottom end **1158a** of the first side plate **1156a** to the second bottom end **1158b** of the second side plate **1156b**. A gap **1162** is formed between the first side plate **1156a** and the second side plate **1156b** as shown in FIG. 30D.

[0394] A first hole **1164a** may be disposed in and extend entirely through the first side plate **1156a** between the first top end **1160a** and the first bottom end **1158a**. A second hole **1164b** may also be disposed in and extend entirely through the second side plate **1156b** between the second top end **1160b** and the second bottom end **1158b**. The second hole **1164b** may be disposed in the second side plate **1156b** at the same location as the first hole **1164a** is disposed in the first side plate **1156a** so that the first hole **1164a** and the second hole **1164b** are substantially aligned. The first and second holes **1164a**, **1164b** may be any size and shape suitable for a shackle or shank of a lock (e.g., shackle **14** of the lock **10**) to pass through. In some embodiments, the first and second holes **1164a**, **1164b** may be imparted with a substantially oval shape. In other embodiments, the first and second holes **1164a**, **1164b** may be imparted with a substantially circular shape.

[0395] The first side **1154a** may further include a first clip support **1166a** on the first top end **1160a**, and the second side **1154b** may include a second clip support **1166b** on the second top end **1160b**. A first vertical support **1168a** may extend from the first clip support **1166a** upward and past the top portion **1110** of the housing **1102**. A second vertical support **1168b** may extend from the second clip support **1166b** upward and past the top portion **1110** of the housing **1102**. In some embodiments, the first vertical support **1168a** and the second vertical support **1168b** may be mirror images of each other.

[0396] A clip connector **1170** may extend between the first vertical support **1168a** and the second vertical support **1168b** thereby connecting the first vertical support **1168a** to the second vertical support **1168b**. In some embodiments, the clip connector **1170** may be imparted with a cylindrical or rod shape.

[0397] The first vertical support **1168a** may have a first front portion **1178a**, and the second vertical support **1168b** may have a second front portion **1178b**. Extending outward from the first front portion **1178a** of the first vertical support **1168a** may be a first guide rail **1180a**, and extending outward from the second front portion **1178b** of the second vertical support **1168b** may be a second guide rail **1180b**. In some embodiments, the first and second guide rails **1180a**, **1180b** may extend along a top flange portion **1176** of the first flange **1124** across the entire depth of the first flange **1124**. In some embodiments, the first and second guide rails **1180a**, **1180b** may be disposed

substantially parallel to one another and may be mirror images of each other.

[0398] The first guide rail **1180a** may have a first end portion **1186a**, opposite the first front portion **1178a** of the first vertical support **1168a**, and the second guide rail **1180b** may have a second end portion **1186b**, opposite the second front portion **1178b** of the second vertical support **1168b**. An end wall **1188** may extend from the first end portion **1186a** of the first guide rail **1180a** to the second end portion **1186b** of the second guide rail **1180b**. The end wall **1188** may connect the first guide rail **1180a** to the second guide rail **1180b**.

[0399] A first flange aperture **1190** may be disposed in and extend entirely through the top flange portion **1125** between the first guide rail **1180a**, the second guide rail **1180b** and the end wall **1188** as shown in FIG. **30A** and **31C**. The first flange aperture **1190** may be imparted with any size suitable to accommodate the hook portion **212** of the clip **200**. In some embodiments, the first flange aperture **1190** has a substantially rectangular shape.

[0400] The clip support unit **1150** may be designed to engage with one or more embodiments of a clip (e.g., **200**, **400**, **800**, **1000**) disclosed herein. In some embodiments, the first and second clip supports **1166a**, **1166b** may be designed to engage or otherwise be in communication with first and second support legs (e.g., **222a**, **222b**), respectively, of a clip (e.g., **200**). The clip connector **1170** may be designed to engage with a coupling member (e.g., **216**) of the clip such that the coupling member can rotate about the clip connector **1170** in order to raise and/or lower a first end (e.g., **204**) of a first elongated member (e.g., **202**) of the clip. In addition, the gap **1162** disposed between the first and second side plates **1156a**, **1156b** of the clip support unit **1150** may be designed to allow a second elongated member (e.g., **228**) of the clip to move in a substantially upwardly or downwardly direction between the first and second side plates **1156a**, **1156b**. The first and second guide rails **1180a**, **1180b** may be designed to engage or otherwise be in communication with first and second rails (e.g., **212a**, **212b**), respectively, of the clip. Furthermore, the first and second holes **1164a**, **1164b** of the clip support unit **1150** may be designed to substantially align with a lock through hole (e.g., **236**) of the clip to enable the shackle of a lock (e.g., shackle **14** of lock **10**) to pass through the housing **1102** and clip and secure same to an EV charge port or an EV charge plug (e.g., **50"**).

[0401] The seventh embodiment of the EV lockout device **1100** may include the clip **200** designed for use therewith. As shown in FIGS. **31A-31C**, the clip **200** may attach to or otherwise engage with the housing **1102**. When in the first clip position, the clip **200** may be placed on the housing **1102** such that the coupling member **216**, the bottom side **210** of the first elongated member **202**, and/or the projection **218** of the clip **200** may engage with the clip connector **1170** of the housing **1102** as best seen in FIG. **31C**. The clip connector **1170** may rest within the nook **217** when in the first clip position, and the coupling member **216** may extend through the divot opening **1122** of the housing **1102** as shown in FIG. **31C**. In addition, when in the first clip position, the first support leg **222a** may engage with the first clip support **1166a** of the clip support unit **1150**, and the second support leg **222b** may engage with the second clip support **1166b** of the clip support unit **1150**. The first elongated member **202** of the clip **200** may extend over the top flange portion **1125** of the first flange **1124**, and the hook portion **212** of the clip **200** may rest within the first flange aperture **1190**. The second elongated member **228** may be disposed within the gap **1162** between the first side **1154a** and the second side **1154b** of the clip support unit **1150** as shown in FIG. **31B**. Further, when in the first clip position, the lock through hole **236** of the clip **200** may substantially align with the first hole **1164a** of the first side **1154a** and the second hole **1164b** of the second side **1154b** of the clip support unit **1150**.

[0402] As discussed above, when in the first clip position, the first and second support legs **222a**, **222b** may engage with the first and second clip supports **1166a**, **1166b**, respectively. In addition, when in the first clip position, the lock through hole **236** of the second elongated member **228** of the clip **200** may substantially align with the first hole **1164a** of the first side **1154a** and the second hole **1164b** of the second side **1154b** of the clip support unit **1150** of the housing **1102**.

[0403] When the EV lockout device **1100** is being placed on or is being removed from the fourth EV charge plug **50''**, the clip **200** may move to a second clip position (not shown). When moving into the second clip position, the first end **204** of the first elongated member **202** may move in a substantially upward direction. In some embodiments, the clip **200** may move to the second clip position either in response to the platform portion **220** being pressed downward so that the first and second support legs **222a**, **222b** compress (e.g., the angle between the platform portion **220** and the first and second support legs **222a**, **222b** decreases). In other embodiments, the clip **200** may move to the second clip position in response to the hook portion **212** of the first elongated member **202** engaging with the first protrusion **64''** of the fourth EV charge plug **50''**. In the second clip position, the first end **204** of the first elongated member **202** may raise up in order for the hook portion **212** to slide over the first protrusion **64''** and into the top opening **53'** of the fourth EV charge plug **50''**. As the first end **204** of the first elongated member **202** raises upward, the coupling member **216** and the projection **218** may move or rotate about the clip connector **1170** of the housing **1102** in a first direction. In some embodiments, as the first end **204** of the first elongated member **202** raises upward, the second elongated member **228** may move in a substantially downward direction such that the space **235** between the bottom end **234** of the second elongated member **228** and the bottom support **1152** of the clip support unit **1150** decreases. In some embodiments, as the second elongated member **228** moves in the substantially downward direction, the lock through hole **236** of the second elongated member **228** may become out of alignment with the first hole **1164a** and the second hole **1164b** of the clip support unit **1150**.

[0404] If the EV lockout device **1100** is being placed onto the fourth EV charge plug **50''**, the clip **200** may move from the first clip position to the second clip position. As the clip **200** is moving from the first clip position to the second clip position, the housing **1102** may engage with the fourth EV charge plug **50''**. In some embodiments, the first flange **1124** of the housing **1102** may engage with the first protrusion **64''** of fourth EV charge plug **50''** such that the first flange **1124** surrounds all or a portion of the first protrusion **64''**.

[0405] After moving into the second clip position in response to the platform portion **220** being depressed, the clip **200** may move back into the first clip position upon release of the platform portion **220**. As the clip **200** moves back into the first clip position, the first end **204** of the first elongated member **202** may lower and the coupling member **216** and the projection **218** may move or rotate about the clip connector **1170** of the housing **1102** in a direction opposite the first direction. In addition, as the clip **200** moves back in the first clip position, the second elongated member **228** may move in a substantially upward direction as the first and second support legs **222a**, **222b** decompress (e.g., the angle between the platform portion **220** and one or more of the first and second support legs **222a**, **222b** increases). If the EV lockout device **1100** has been placed on the fourth EV charge plug **50''**, the hook portion **212** may be disposed in one or more of the first flange aperture **1190** of the housing **1102** and the top opening **53'** of the fourth EV charge plug **50''**, when the clip **200** returns to the first clip position. Additionally, when the clip **200** moves back into the first clip position, the first hole **1164a** of the clip support unit **1150**, the lock through hole **236** of the second elongated member **228**, and the second hole **1164b** of the clip support unit **1150** may be substantially aligned such that the shackle **14** of lock **10** may be threaded through the first hole **1164a**, the lock through hole **236**, and the second hole **1164b**, or vice versa. The shackle **14** may then be securely attached to the body **12** of the lock **10**, thereby placing the EV lockout device **1100** in a locked position as shown in FIG. 32.

[0406] When in the locked position, the clip **200** cannot fully move to the second clip position (e.g., the hook portion **212** cannot be raised high enough to slide out of the top opening **53'** and/or the first flange aperture **1190** and back over the first protrusion **64''** of the fourth EV charge plug **50''**) as the shackle **14** prevents the second elongated member **228** of the clip **200** from moving sufficiently downward. To remove the EV lockout device **1100** from the fourth EV charge plug **50''**, the shackle **14** of the lock **10** may be disengaged from the body **12** of the lock **10** and removed

from the clip support unit **1150** of the housing **1102** by threading the shackle **14** through the second hole **1164b**, the lock through hole **236**, and the first hole **1164a**, or vice versa. Once the lock **10** is removed from the EV lockout device **500**, the platform portion **220** of the clip **200** may be depressed causing the clip **200** to move from the first clip position to the second clip position in which the first and second support legs **222a**, **222b** compress and the hook portion **212** is raised. The hook portion **212** may then be moved over and away from the top opening **53'** of the fourth EV charge plug **50''** and the first flange aperture **1190** as the EV lockout device **1100** is slid off or otherwise removed from the fourth EV charge plug **50''** (e.g., the first flange **1124** is removed from the first protrusion **64'''**).

[0407] FIG. **33** illustrates a method **1200** of using the first, second, third, fourth, fifth, and sixth embodiments of the EV lockout device (e.g., **100**, **300**, **500**, **600**, **700**, **900**, **1100**, respectively) discussed above.

[0408] At a step **1202**, a clip (e.g., **200**, **400**, **800**, **1000**) may be attached to a housing (e.g., **102**, **302**, **502**, **602**, **702**, **902**, **1102**) of the EV lockout device. As discussed above, the clip may be attached to the housing by engaging a coupling member (**216**, **416**, **816**, **1016**) and a projection (**218**, **418**, **818**, **1018**) of the clip with a clip connector (**170**, **370**, **570**, **670**, **770**, **970**, **1170**) of the housing such that the clip connector rests within a nook (**217**, **417**, **817**, **1017**) of the clip.

[0409] Once attached to the housing, the clip may be disposed be in a first clip position. As discussed above, when in the first clip position, first and second support legs (e.g., **222a**, **222b**; **422a**, **422b**; **822a**, **822b**; **1022a**, **1022b**) of the clip may engage with first and second clip supports (e.g., **166a**, **166b**; **366a**, **366b**; **566a**, **566b**; **666a**, **666b**; **766a**, **766b**; **966a**, **966b**, **1166a**, **1166b**), respectively, of the housing or the bottom support unit (see, e.g., **951**) of the housing (e.g., **902**). The first and second support legs may be in a neutral position, when the clip is in the first clip position, such that the first and second support legs are not compressed. In addition, a first elongated member (e.g., **202**, **402**, **802**, **1002**) of the clip may extend over a top flange portion (e.g., **125**, **325**, **525**, **625**, **725**, **1125**) of a first flange (e.g., **124**, **324**, **524**, **624**, **724**, **1124**) or an inner side (e.g., **931**) of a first flange (e.g., **924**) of the housing when the clip is in the first clip position. And, a second elongated member (e.g., **228**, **428**, **828**, **1028**) of the clip may be disposed within a gap (e.g., **162**, **362**, **562**, **662**, **762**, **962**, **1162**) between first and second sides (e.g., **154a**, **154b**; **354a**, **354b**; **554a**, **554b**, **654a**, **654b**; **754a**, **754b**, **954a**, **954b**; **1154a**, **1154b**) of a clip support unit (e.g., **150**, **350**, **550**, **650**, **750**, **950**, **1150**) when the clip is in the first clip position. Furthermore, first and second holes (e.g., **164a**, **164b**; **364a**, **364b**; **564a**, **564b**; **664a**, **664b**; **764a**, **764b**; **964a**, **964b**, **1164a**, **1164b**) of the clip support unit of the housing may be in substantial alignment with a lock through hole (e.g., **236**, **436**, **836**, **1036**) of the second elongated member of the clip.

[0410] Once the clip is attached to the housing, the EV lockout device may be placed onto or into a port housing (e.g., **22**, **22'**, **22''**) of an EV charge port (e.g., **20**, **20'**, or **20''**) or a plug housing (e.g., **52**, **52'**, **52''**, **52'''**) of an EV charge plug (e.g., **50**, **50'**, **50''**, **50'''**).

[0411] At a step **1204**, a platform portion (e.g., **220**, **420**, **820**, **1020**) of the clip may be pressed downward causing the first and second support legs of the clip to compress, which in turn raises a first end (e.g., **204**, **404**, **804**, **1004**) of the first elongated member of the clip, thereby moving the first clip from the first clip position to a second clip position.

[0412] When in the second clip position, the first end of the first elongated member of the clip may engage with a locking mechanism (e.g., **30**, **30'**, **30''**) of the EV charge port or a locking mechanism (e.g., **54**, **54'**, **54''**, **54'''**) of the EV charge plug at a step **1206**. In some embodiments, the first end of the clip (e.g., **204**, **804**, **1004**) engages with the locking mechanism (e.g., **30**, **30'**, **30''**) of the EV charge port (e.g., **20**, **20'**, **20''**) or the locking mechanism (e.g., **54''**, **54'''**) of the EV charge plug (e.g., **50''**, **50'''**) by raising up in order to hook onto or into the locking mechanism. In other embodiments, the first end of the clip (e.g., **404**) engages with the locking mechanism of the EV charge plug (e.g., **50**, **50'**) by raising up to allow a locking member (e.g., **56**, **56'**) of the locking mechanism (e.g., **54**, **54'**) to latch onto an inner side (e.g., **385**, **685**) of a center divider (e.g., **384**,

684) of the housing (e.g., **302, 602**) of the EV lockout device. In addition, when in the second clip position, the lock through hole of the second elongated member of the clip may be out of substantial alignment with the first and second holes of the clip support unit of the housing.

[0413] At a step **1208**, the first flange of the housing of the EV lockout device may engage with a first portion of the port housing of the EV charge port or a first portion of the plug housing of the EV charge plug while the clip is in the second clip position. In some embodiments, the first portion of the EV charge port may be provided in the form of a first protrusion (e.g., **34, 34'**). In some embodiments, the first portion of the EV charge port may be provided in the form of a locking mechanism aperture (e.g., **24''**). In some embodiments, the first portion of the EV charge plug may be provided in the form of a first protrusion (e.g., **64, 64', 64'', 64'''**). In some embodiments, the first portion of the port housing of the EV charge port (e.g., **20, 20'**) may include a channel (e.g., **44, 44'**). In some embodiments, the first portion of the plug housing of the EV charge plug (e.g., **50, 50'**) may include a protrusion ridge (e.g., **62, 62**). In some embodiments, a ridge (e.g., **128, 528**) may extend from a bottom flange portion (e.g., **127, 527**) of the first flange (e.g., **124, 524**) and may engage with the corresponding channel (e.g., **44, 44'**) of the EV charge port as the first flange is mating with the first portion of the EV charge port. In some embodiments, a flange channel (e.g., **328, 628**) may be disposed in a bottom flange portion (e.g., **327, 627**) of the first flange (e.g., **324, 624**) and may engage with the corresponding protrusion ridge (e.g., **62, 62**) of the EV charge plug as the first flange is mating with the first portion of the plug housing of the EV charge plug.

[0414] In some embodiments, the housing (e.g., **102, 302**) of the EV lockout device may have a second flange (e.g., **126, 326**). If the housing of the EV lockout device has a second flange, then the second flange may be mated with a second portion of the port housing of the EV charge port or the plug housing of a second portion of a EV charge plug at an optional step **1210**. In some embodiments, the second portion of the port housing of the EV charge port may be provided in the form of a second protrusion (e.g., **38**) and the second portion of the plug housing of the EV charge plug may be provided in the form of a second protrusion (e.g., **68**). In some embodiments, the first flange and the second flange of the EV lockout device may be mated with the first portion and the second portion of the EV port housing of the EV charge port or the plug housing of the EV charge plug at the same time. In other embodiments, the clip may engage with the locking mechanism of the EV charge port or the EV charge plug as the first flange is mating with the first protrusion and the second flange, where applicable, is mating with the second portion of the EV charge port or the EV charge plug.

[0415] At a step **1212**, once the clip has engaged with the locking mechanism of the EV charge port or EV charge plug, the first flange (including any ridges or flange channels) and the second flange, where applicable, have mated with the first portion and the second portion, respectively, of the EV charge port or the EV charge plug, the platform portion of the clip may be released causing the clip to move from the second clip position back to the first clip position. As the clip moves from the second clip position to the first clip position, the first and second support legs of the clip decompress and the first end of the first elongated member of the clip lowers. In some embodiments, when the clip moves back into the first clip position after being placed onto or into the EV charge port or the EV charge plug, the first end of the first elongated member of the clip (e.g., **200**) may be disposed on an inner side (e.g., **33, 33'**) of a locking member (e.g., **32, 32'**) of the locking mechanism (e.g., **30, 30'**) of the EV charge port (e.g., **20, 20'**). In other embodiments, the first end of the first elongated member of the clip (e.g., **1000**) may be disposed in a bottom opening (e.g., **35**) of the locking mechanism (e.g., **30''**) of the EV charge port (e.g., **20''**). In some embodiments, the first end of the first elongated member of the clip (e.g., **800, 200**) may be disposed in a top opening (e.g., **53, 53**) of the locking mechanism (e.g., **54'', 54'''**) of the EV charge plug (e.g., **50'', 50'''**) when the clip returns to the first clip position. In further embodiments, when the clip (e.g., **400**) moves back into the first clip position, the locking member (e.g., **56, 56'**) of the locking mechanism (e.g., **54, 54'**) of the EV charge plug (e.g., **50, 50'**) may be disposed on the

inner side (e.g., **385, 685**) of the center divider (e.g., **384, 684**) of the housing (e.g., **300, 600**).

[0416] At a step **1214**, a lock **10** may be attached to the EV lockout device once the clip is back in the first clip position. When the clip returns to the first clip position after being placed on the EV charge port or EV charge plug, the first and second holes of the clip support unit of the housing may be in substantial alignment with the lock through hole of the second elongated member of the clip. A shackle **14** of the lock may be threaded through the first hole of the clip support unit, the lock through hole of the second elongated member, and the second hole of the clip support unit, or vice versa, and then securely reattached or locked to the body **12** of the lock **10**.

[0417] To remove the EV lockout device from the EV charge port or EV charge plug, the lock **10** may be removed at a step **1216**. The lock **10** may be removed, by detaching one or both ends (e.g., **16, 18**) of the shackle **14** of the lock **10** and threading the shackle **14** back through the first hole of the housing, the lock through hole of the clip, and the second hole of the housing, or vice versa. Once the lock **10** is removed from the EV lockout device, the EV lockout device can be removed from the EV charge port or EV charge plug.

[0418] At a step **1218**, the platform portion of the clip can be depressed causing the clip to move from the first clip position to the second clip position where the first end of the first elongated member of the clip is raised up. Once raised, the first end of the clip can be disengaged from the locking mechanism of the EV charge port or the EV charge plug at a step **1220**.

[0419] At a step **1222**, the first flange can be removed from the first portion of the EV charge port or EV charge plug by pulling the first flange off of the first portion of the EV charge port or EV charge plug.

[0420] If the housing has a second flange, then the second flange can be removed from the EV charge port or EV charge plug by pulling the second flange off of the second portion of the EV charge port or EV charge plug at the optional step **1224**. In some embodiments, the first and second flanges are removed from the first and second portions of the EV charge port or plug at the same time. In some embodiments, the clip is disengaged from the housing at the same time as the first flange is removed from the first portion and the second flange, where applicable, is removed from the second portion of the EV charge port or EV charge plug.

[0421] At a step **1226**, once the EV lockout device has been removed from the EV charge port or EV charge plug, the platform portion of the clip can be released and the clip returned to the first clip position.

[0422] It will be appreciated by those skilled in the art that while the above disclosure has been described above in connection with particular embodiments and examples, the above disclosure is not necessarily so limited, and that numerous other embodiments, examples, uses, modifications and departures from the embodiments, examples and uses are intended to be encompassed by the claims attached hereto. The entire disclosure of each patent and publication cited herein is incorporated by reference, as if each such patent or publication were individually incorporated by reference herein. Various features and advantages of the above disclosure are set forth in the following claims.

Claims

1. A housing for an electric vehicle (EV) lockout device comprising: a base having a front side and a back side; a first flange extending from the front side of the base, the first flange configured to engage with a first portion of a port housing of an EV charge port or a first portion of a plug housing of an EV charge plug; and a clip support unit extending from the back side of the base, the clip support unit comprising: a first side plate having a first top end, a first bottom end opposite the first top end, and a first opening disposed between the first top end and the first bottom end; and a second side plate having a second top end, a second bottom end opposite the second top end, and a first opening disposed between the second top end and the second bottom end, wherein the second

side plate is disposed substantially parallel to the first side plate, and wherein the second opening is substantially aligned with the first opening.

2. The housing for an EV lockout device of claim 1, wherein the first opening and the second opening have the same shape and size.

3. The housing for an EV lockout device of claim 1, wherein the first flange has a bottom flange portion, the housing for an EV lockout device further comprising a ridge that extends from the bottom flange portion of the first flange, wherein the ridge is configured to engage with a channel in the first portion of the port housing of the EV charge port.

4. The housing for an EV lockout device of claim 1, wherein the first flange has a bottom flange portion, the housing for the EV lockout device further comprising a flange channel that extends through the bottom flange portion of the first flange, wherein the flange channel is configured to engage with a protrusion ridge of the plug housing of the EV charge plug.

5. The housing for an EV lockout device of claim 1, further comprising a first vertical member disposed on the back side of the base and a second vertical member disposed on the back side of the base, wherein the first vertical member is disposed substantially parallel to the second vertical member.

6. The housing for an EV lockout device of claim 5, further comprising a clip connector that extends between the first vertical member and the second vertical member.

7. The housing for an EV lockout device of claim 5, further comprising a first support member connected to the first top end of the first side plate and a second support member connected to the second top end of the second side plate, wherein the first vertical member extends upwardly from the first support member and the second vertical member extends upwardly from the second support member.

8. The housing for an EV lockout device of claim 1, further comprising a second flange extending from the front side of the base, wherein the second flange is disposed below the first flange, and wherein the second flange is configured to engage with a second portion of the port housing of the EV charge port or a second portion of the plug housing of the EV charge plug.

9. The housing for an EV lockout device of claim 8, wherein a ridge is disposed between the first flange and the second flange, wherein the ridge is configured to engage with a channel in the first portion of the port housing of the EV charge port.

10. The housing for an EV lockout device of claim 8, wherein a flange channel is disposed between the first flange and the second flange, wherein the flange channel is configured to engage with a protrusion ridge of the first portion of the plug housing of the EV charge plug.

11. A clip for an electric vehicle (EV) lockout device comprising: a first elongated member having a first end and a second end; a second elongated member having a top portion and a bottom portion opposite the top portion, wherein the second elongated member is disposed substantially perpendicular to the first elongated member; a platform portion connected to the second end of the first elongated member; a first support leg extending from the platform portion; and a second support leg extending from the platform portion, wherein the second support leg is disposed substantially parallel to the first support leg.

12. The clip for an EV lockout device of claim 10, wherein the second elongated member extends downwardly from the platform portion.

13. The clip for an EV lockout device of claim 10, wherein a hole is disposed between the top portion and the bottom portion of the second elongated member.

14. The clip for an EV lockout device of claim 10, wherein the first support leg and the second support leg each extend downwardly from the platform portion at an angle.

15. The clip for an EV lockout device of claim 14, wherein the first support leg and the second support leg are configured to compress such that the angle between the platform portion and first support leg or the second support leg decreases.

16. The clip for an EV lockout device of claim 14, the angle is about 45 degrees.

17. The clip for an EV lockout device of claim 10, further comprising a hook portion that extends downwardly from the first end of the first elongated member, wherein the hook portion is configured to engage with a locking mechanism of an EV charge port or an EV charge plug.

18. The clip for an EV lockout device of claim 10, wherein the first elongated member has a first side and a second side substantially parallel to the first side, the first side and the second side extending between the first end and the second end of the first elongated member.

19. The clip for an EV lockout device of claim 18, wherein a first guide member extends downwardly from the first side of the first elongated member and a second guide member extends downwardly for the second side of the first elongated member.

20. A method of using an electric vehicle (EV) lockout device, the EV lockout device comprising a clip and a housing, the clip comprising a first elongated member having a first end and a second end, opposite the first end, a second elongated member disposed perpendicular to the first elongated member, wherein a hole is disposed on the second elongated member, and a platform portion connected to the second end of the first elongated member, the method comprising the steps of: attaching the clip to the housing, wherein the first elongated member extends over a first flange of the housing, and the hole is in substantial alignment with a first opening of the housing and a second opening of the housing; depressing the platform portion of the clip to move the clip from a first clip position to a second clip position, wherein the clip is in a neutral position when in the first clip position, and wherein the first end of the first elongated member is raised in the second clip position; engaging the first end of the clip with a locking mechanism of an EV charge port or an EV charge plug; mating the first flange of the housing with a first portion of the port housing of the EV charge port or a first portion of a plug housing of the EV charge plug; releasing the platform portion of the clip to move the clip from the second clip position to the first clip position; attaching a lock to the EV lockout device by threading a shackle of the lock through the first opening of the housing, the hole of the second elongated member of the clip, and the second opening of the housing, and attaching the shackle to a body of the lock.
